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Title and Front Matter

The Space Omics and Medical Atlas (SOMA) and international astronaut biobank

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Summary

Spaceflight induces molecular, cellular, and physiological shifts in astronauts and poses myriad biomedical challenges to the human body, which are becoming increasingly relevant as more

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humans venture into space[1-6]. Yet, current frameworks for aerospace medicine are nascent and lag far behind advancements in precision medicine on Earth, underscoring the need for rapid development of space medicine databases, tools, and protocols. Here, we present the Space Omics and Medical Atlas (SOMA), an integrated data and sample repository for clinical, cellular, and multi-omic research profiles from a diverse range of missions, including the NASA Twins Study[7], JAXA CFE study[8,9], SpaceX Inspiration4 crew[10-12], plus Axiom and Polaris. The SOMA resource represents a >10-fold increase in publicly available human space omics data, with matched samples available from the Cornell Aerospace Medicine Biobank. The Atlas includes extensive molecular and physiological profiles encompassing genomics, epigenomics, transcriptomics, proteomics, metabolomics, and microbiome data sets, which reveal some consistent features across missions, including cytokine shifts, telomere elongation, and gene expression changes, as well as mission-specific molecular responses and links to orthologous, tissue-specific murine data sets. Leveraging the datasets, tools, and resources in SOMA can help accelerate precision aerospace medicine, bringing needed health monitoring, risk mitigation, and countermeasures data for upcoming lunar, Mars, and exploration-class missions.

Main Text

With an exponential increase in launches since 2019, space is rapidly becoming more accessible[1]. Multiple commercial and state-sponsored groups are developing roadmaps to construct space stations, moon bases, Mars colonies, and other permanent establishments beyond Earth[1]. While innovation across the aerospace sector makes these ambitions technologically achievable, the biomedical challenges for crews in these extraplanetary habitats still need to be addressed, as humans did not evolve to survive in such extreme environments. The clinical consequences of this evolutionary mismatch of spaceflight exposure and adaptation have revealed a plethora of challenges to long-term space habitation, including a loss in bone density and muscle mass[2], spaceflight-associated neuro-ocular syndrome (SANS)[3], perturbed immune function[4], and spaceflight anemia[5]. These physiological changes during spaceflight appear to imprint on the health status in humans; a chief example of this long term effect is the increased risk of cardiovascular pathology observed in astronauts compared to age-matched controls[6].

Before long-term space habitation is feasible, these biomedical challenges must be mitigated. However, the etiologies driving them are not understood, with the low number of astronauts yielding limited opportunities for in-depth biomolecular characterization. For example, two of the largest multi-omic studies to date were the NASA Twins Study, which published an in-depth molecular and cognitive profile of a single astronaut[7], and the Japanese Aerospace Exploration Agency (JAXA) Cell-Free Epigenome (CFE) project, which profiled cfDNA and cfRNA in six astronauts[8,9]. Thus, achieving statistical power requires integrating data from other cohorts, such as the MARROW study, and other missions[5,10]. Further, these analyses are complicated by the substantial variation in physiologic responses to spaceflight among astronauts. Therefore, there is

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a need for increasingly large, detailed multi-omic profiles of astronauts to characterize the diversity of physiologic shifts as a function of spaceflight[11].

To achieve this end, we have leveraged the burgeoning commercial spaceflight industry. With the launch of SpaceX's 2021 Inspiration4 (I4) mission, a cohort of all-civilian astronauts successfully completed a high elevation (590km), three-day orbital mission within a SpaceX Dragon capsule. Using recently developed protocols, the crew participated in a range of biospecimen collections before, during, and after their mission[12]. We used the I4 biospecimens to deeply profile the impact of the stressors of spaceflight (e.g., microgravity and space radiation) on crew physiology and health. We also compared these results to prior missions and control datasets, creating the largest-to-date molecular atlas of the impact of spaceflight on the human body, encompassing almost 3,000 samples and over 75 billion sequenced nucleic acids. Collectively, these resources are referred to as the Space Omics and Medical Atlas (SOMA), and the samples are linked to a Cornell Aerospace Medicine Biobank (CAMbank) that stores viably frozen specimens for future, additional analyses.

As with Earth-based cohorts[13], these accessible data – when profiled and aggregated at scale – will enable the development of both personalized and general medical guidance for astronauts[14]. A large group of subject matter experts in artificial intelligence recently released recommendations focused on the importance of generating and archiving space data into the NASA Open Science Data Repository (OSDR)[15,16] to enable autonomous and intelligent precision space health systems, and to monitor, aggregate and assess biomedical statuses for future deep space missions[17]. Additionally, the study of the parallels between the physiological impacts of spaceflight and aging, chronic disease and immune system disorders using omics data can pave the way for therapeutics applicable to conditions on Earth.

Here, we present a detailed guide to the SOMA resource, which includes the 2,911 samples collected during the I4 mission[11,12], as well as spatial transcriptomics data, long-read profiles of astronaut RNA, microbiome data, exosome profiles, and in-depth immune diversity maps. Additional spaceflight data were annotated and compiled into the SOMA portal to help contextualize gene, protein, or metabolite dynamics, including as well as data from the NASA Twins Study[7], JAXA's CFE mission[8,9], single-cell RNA-seq data after simulated microgravity on PBMCs[18], and mouse spaceflight data matched to human orthologs. In addition to rigorous dataset annotations, we detail (1) a comparison on NASA Twins Study conclusions and flight dynamics comparing short- and long-duration missions, (2) cell-type specific responses to spaceflight previously undocumented in astronauts, (3) cell-free RNA (cfRNA) expression profiles showing hematological responses during recovery from spaceflight, and (4) additional analyses on individual responses to spaceflight from proteomic, transcriptomic, and microbiome data. Data and samples generated in this study are available through the Space Omics and Medical Atlas (https://soma.weill.cornell.edu), NASA Open Science Data Repository

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(https://osdr.nasa.gov/bio/) and CAMbank (https://cambank.weill.cornell.edu/), which offer an unprecedented view of the multi-system omics changes before, during, and after spaceflight.

Results

Comprehensive astronaut data resource

To generate a comprehensive profile of the physiological changes of the I4 crew (ages 29, 38, 42, 51), thirteen biospecimen sample types were collected and processed, including whole blood, serum, Peripheral Blood Mononuclear Cells (PBMCs), plasma, extracellular vesicles and particles (EVPs) derived from plasma, dried blood spots, oral swabs, nasal swabs, skin biopsies, skin swabs, capsule (SpaceX Dragon) swabs, urine, and stool specimens[12]. After collection, samples were subject to a battery of multi-omic assays, including clinical (CLIA) whole genome sequencing (WGS), a clonal hematopoiesis panel, direct RNA-sequencing (dRNA-seq), single-nuclei RNA-seq (snRNA-seq), single-nuclei ATAC-seq (snATAC-seq), single-cell B-cell repertoire (BCR) and T-cell repertoire (TCR) V(D)J sequencing, untargeted plasma proteomics (LC-MS/MS), untargeted plasma metabolomics, cell-free DNA (cfDNA) sequencing, cell-free RNA (cfRNA), metagenomics, metatranscriptomics, and spatially resolved transcriptomics. Additionally, chemokine/cytokine and cardiovascular biomarkers were quantified and a CLIA lab (Quest Diagnostics) was used to perform a complete blood count (CBC) and comprehensive metabolic panel (CMP) (**Fig 1a**). Datasets were generated across ten timepoints: three pre-flight (L-92, L-44, L-3), three in-flight (Flight Day (FD) 1, FD2, FD3), one immediately post-flight (R+1) and three recovery (R+45, R+82, R+194) time points (**Fig 1b**), spanning 289 days. Assays were performed on all crew members unless otherwise noted (**Fig 1b, Supplementary Table 1**).

A total of 2,911 samples were banked, with 1,194 samples processed for sequencing, imaging, and biochemical analysis (**Supplementary Table 1**). These results and assays subsume and expand upon work and protocols from prior missions, including the JAXA CFE study, the NASA Twins Study, and some NASA astronauts (**Fig 1c**). This latter category spans studies primarily from the ISS that lack certain metadata, primarily duration spent in space and launch dates, in order to maintain astronaut anonymity. These studies include chemokine/cytokine biomarker panels (n=46 astronauts), comprehensive metabolic panels, telomere length qPCR, and ISS-surface metagenomic profiling (**Fig 1c**).

The SOMA resources were first compared to the NASA OSDR database, which contains all publicly accessible human omics data from spaceflight and ground analog studies. OSDR hosts 76 human omics studies, of which 11 are from human primary cells exposed to spaceflight. The other studies encompassed cell line and ground studies, including high-altitude studies (**Extended Data Fig 1a, Supplementary Table 2**), which were all merged with the SOMA data set. Once merged, the total number of sequenced nucleic acid molecules from this study

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represents a >10-fold increase in the total amount of human omics data in OSDR (**Extended Data Fig 1b)**, across all spaceflight studies, ground studies, cell line, and primary cell experiments (**Supplementary Table 2, Supplementary Table 3, Extended Data Fig 1**).

The data from the missions were then divided into three analysis timeframes: 1) flight profiles (FP), 2) recovery profiles (RP), and 3) longitudinal profiles (LP) (**Extended Data Fig 1c**). Flight profiles reveal the most immediate impact of spaceflight, recovery profiles catalog changes that occur after return to Earth, and the longitudinal profiles identify changes that have not returned to baseline after returning to Earth. We focused on several outputs for the resource, including first calculating differentially expressed genes (DEGs) for (1) PBMC snRNA-seq, (2) whole blood dRNA-seq, (3) skin spatially-resolved transcriptomics, and (4) cfRNA. We also mapped differentially methylated genes from whole blood dRNA-seq, differentially accessible regions (DARs) from PBMC snATAC-seq, isotype identification from TCR and BCR V(D)J sequencing, differentially abundant proteins (DAPs) from plasma and EVP proteomics, differential metabolites from LC-MS metabolomics, and microbial differentials from metagenomic and metatranscriptomic assays (**Extended Data Fig 2**), with all raw and processed data annotated in OSDR (**Supplementary Table 4**).

Inspiration4 reproduces NASA Twins Study

Telomere elongation was previously described in three astronauts who stayed for six months to one year aboard the ISS[7,19,20], but it was unclear how quickly such a phenotype appeared in astronauts. Interestingly, average telomere length in all Inspiration4 crew members increased during spaceflight (17-22% longer), and this trend was statistically significant (mixed effects linear model: p=0.0048) (**Fig 2a**). This finding is particularly notable, given the shorter mission duration (3 total days) and higher elevation of the I4 mission compared to the ISS studies, indicating that telomere length dynamics respond much more rapidly to spaceflight than previously observed.

We then compared the DEGs and cytokine changes from the Twins Study to those observed in the snRNA-seq data from I4 mission, as well as compared to the expected DEGs of the assay from replicate negative control donor PBMCs (see methods). The cross-mission DEG comparison highlighted a consistent response between both types of T-cells, including CD4+ and CD8+ markers (552 DEGs, and 608 DEGs, respectively, both $<2.2 \times 10^{-16}$), across both sorted T-cells or single-cell annotated cells (**Fig 2b**). Conversely, B-cells were less responsive to spaceflight, as expected from prior work in the Twins Study[7], which showed B-cells as either not significant or less responsive to spaceflight. For the four overlapping cytokines measured in the Twins Study with our panel, we found significant increases in three: IL6 ($p = 0.014$), IL10 ($p = 0.021$), and CCL2 ($p = 0.040$) (**Fig 2c**), and these cytokines also showed changes and similar increases in other long-duration (>6 month) crews (**Fig 2c**). However, we also ran a differential

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analysis of all cytokines detected on the I4 mission, to detect any differences from the Twins Study. Indeed, BDNF showed a statistically significant decrease ($p = 0.00011$, $q = 0.0153$) and IL19 showed a statistically significant increase ($p = 0.00015$, $q = 0.0153$) during the post-flight (R+1) timepoint that returned to baseline during recovery (R+45, R+82) (**Fig 2d**).

Distinct RNA fingerprints of spaceflight

Beyond recapitulating known biomarkers of spaceflight, the Atlas integrated newer assays that were not available in prior missions, with a particular emphasis on RNA profiling. The first novel assay was spatially-resolved transcriptomics on skin biopsies, which were obtained from all crew members during one pre-flight time point (L-44) and the day after landing back on Earth (R+1). The 4mm biopsies were stained with markers for DNA, PanCK, FAP, and aSMA and then processed with the NanoString GeoMx Digital Spatial Profiler (DSP), where regions of interest (ROIs) were selected based on the tissue structures identified by the fluorescence staining (**Extended Data Fig 3a**). After filtering out outliers, the RNA counts were used for downstream data analysis, generating 95 ROIs across four skin compartments: outer epidermis (OE), inner epidermis (IE), outer dermis (OD), and vasculature (VA) (**Extended Data Fig 3b,c**), and which revealed a distinct set of DEGs, including JAK-STAT signaling, and melanocyte signatures (**Fig 3**).

A second RNA assay for spaceflight integrated into SOMA was cfRNA profiling, which has recently been established as a dynamic tool for mapping temporal alterations in cfRNA composition and cell lysis[21]. However, bulk cfRNA had not been utilized to measure the response of spaceflight until the JAXA CFE study[8,9] and the I4 mission[12]. Using PCA, we identified a distinct separation in cfRNA profiles pre-flight vs post-flight/recovery for I4, suggesting a systemic physiological shift likely induced by space travel (**Extended Data Fig 3d**). This was further reflected in the differential abundance of cfRNA genes across various time points, revealing specific patterns of noncoding expression (**Extended Data Fig 3e)** and RNA types (**Extended Data Fig 3f**) that correspond with the spaceflight timeline. The cell-type proportions inferred from the cfRNA profiles also exhibited spaceflight-associated variation over time, and also showed variation distinct from a set of healthy blood donor controls (n=35) (**Fig 3a, Supplementary Table 5**). Cell types that showed significant post-flight shifts in proportion included hepatocytes, kidney endothelial cells, hematopoietic stem cells, and melanocytes (**Fig 3b, Supplementary Table 5**). Notably, the melanocyte cell proportions that demonstrated significant changes post-flight (**Fig 3c**) were also found in the spatial skin transcriptomics data, providing additional evidence of adaptive skin responses to the space environment.

A third novel RNA method applied to these spaceflight samples focused on RNA isoforms and RNA modifications (epitranscriptome), though direct RNA sequencing (DRS) on the Oxford Nanopore Technologies (ONT) PromethION, and deep RNA-sequencing (>400M reads/sample)

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on the Ultima Genomics UG100. These data quantified genes that were differentially expressed and displayed differential m^6A methylation (**Extended Data Fig 4a,b**), or both, and were analyzed for enriched GO pathways. We identified a set of sites (**Extended Data Fig 4b;** Set M-I) that undergo hypomethylation during recovery, another set (Set M-II) that are detectably hypermethylated from spaceflight, and a set (Set M-III) that exhibits novel hypermethylation during recovery and longitudinally as well. The common pathways in all three M-I/II/III sets (**Extended Data Fig 4b)** showed evidence of radiation and telomere response, including "TSAI response to radiation therapy" and "Wiemann telomere shortening"[22]. In addition, the set of downregulated pathways after landing (recovery) were distinct, including genes associated with breathing regulation (e.g. CO_2 and O_2 takeup and release by erythrocytes), which matches those pathways associated with crew response ranges (below, **Fig 5**). Although further studies are needed to validate and delineate the potential mechanisms of these RNA dynamics (cfRNA, spatial, and RNA modifications), these pathways suggest a potential relationship between RNA expression and methylation in regulating hematological and dermatological functions upon return to Earth.

Gene regulatory changes during recovery

Leveraging the time-series data, we next analyzed PBMC gene expression from snRNA-seq to discern if unique DEGs are present at each time point. We examined DEGs from immediately postflight (FP1) through the recovery profiles (RP1 and RP2) to observe how gene expression profiles are re-established after spaceflight. The number of DEGs was used to quantify the severity and recovery of crew response, and we also compared identified DEGs from a negative control group, including two healthy donors and read-depth/cell-count matched permutation group (**Extended Data Fig 5**), with an average of >700 cells/crew member, per timepoint, per cell type. The DEG count (adj p-val < 0.05, |log2FC| > 0.5) decreased from FP1 to RP1 in CD16+ monocytes, dendritic cells, natural killer (NK) cells, 'other' T cells (not CD4+ or CD8+), uncategorized ('other') cells, and in pseudobulk (calculated from additive counts across cell types to represent PBMCs). However, they increased in B cells, CD4+ T cells, CD8+ T cells, and CD14+ monocytes (**Fig 4a**); by RP2, all cell types had lower DEG counts than at FP1 and began to approach the expected noise range of single-cell DEGs (**Fig 4a**).

Since DEG counts were higher at RP1 than FP1 for several cell types, we hypothesized the introduction of distinct DEGs after landing back on Earth. First we calculated the percentage of DEGs unique to FP1 at both RP1 and RP2, focusing on cell-type specific DEGs (**Fig 4b**). By LP2 ([L-92, L-44, L-3] vs [R+82]), we observed that nearly all differential gene expression was driven by the same set of genes that were differentially expressed at FP1, with the exception of CD14+ monocytes, which had a unique longitudinal profile not strongly connected with the DEG responses in other cells. With the exception of CD14+ monocytes, the RP1 profile is distinct in the abundance of DEGs not present in the FP1, but this perturbation almost entirely

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(>95% of genes) disappeared after flight (RP2 and LP2). From the DEGs shared with FP1, we observed that gene expression directionality reversed for nearly all DEGs between FP1 and each recovery profile (**Fig 4c**), indicating a return to baseline for those differentially expressed genes. The exception to this were two genes from the CD14+ monocyte population, AHR (FP1: log2FC 0.742, RP1: log2FC 0.808) and PELI1 (FP1: log2FC 0.539, RP1: log2FC 0.560) (**Fig 4c)**, in which there was a positive log2FC in expression at both FP1 and RP1.

From the DEGs that were unique to FP1, we next quantified the uniformity of the pathway enrichment between two PBMC lineages: T cells (CD4+, CD8+) and monocytes (CD14+, CD16+) (**Supplementary Table 6**). Of note, 211 (70.8%) genes were unique to individual cell types and 87 (29.2%) were shared between two or more cell types (**Extended Data Fig 6a**). To identify enriched pathways among each cell type's gene set, over-representation analysis was performed on KEGG pathways (**Supplementary Table 6)**. Even though 70.8% of genes were unique to each cell type, their response was more convergent; unique pathways only ranged from 4.2% (CD4+ T cells) to 42.5% (CD14+ monocytes) (**Extended Data Fig 6b**). Yet, some pathways were cell-type specific, such as the circadian entrainment pathway in CD14+ monocytes, with genes GNG2, GNAS, PRKCB, CREB1, and CAMK2D driving the overrepresentation from this pathway's gene set (**Fig 4d**). Also, our data suggest that overrepresented pathways are sometimes more associated with cell lineage (e.g. all T-cells) than with the individual cell type (e.g. CD4). This is evident from the Longevity Regulating pathway that shares the PRKACB, NFKB1, and PIK3CA genes between both CD4+ and CD8+ T cell populations (**Fig 4d**).

Additionally, we found that certain pathways can be enriched across specific cell types belonging to different lineages. The inflammatory mediator regulation of the "TRP channels" pathway, for example, was overrepresented in both CD8+ T cells and CD14+ monocytes, with each cell type contributing different gene sets associated with the pathway (**Fig 4d**). Other pathways, such as the IL-17 signaling pathway, contain mixtures of shared genes and unique genes and are significantly overrepresented across all cell types (**Fig 4d**). This also indicates that while cell types are distinct, they appear to have a set of core pathways in response to spaceflight, with 30-60% of overrepresented pathways shared both within and between lineages (**Extended Data Fig 6b**).

To examine the chromatin accessibility and regulatory landscape in each cell population, we then analyzed the transcription factor binding sites (TFBS) motif accessibility changes from snATAC-seq data to identify top motifs in flight and recovery profiles (**Extended Data Fig 7**). We first observed that increased gene expression was correlated with more accessibility at the transcription start sites (TSSs), and closed chromatin was associated with down-regulated genes (all p-values <0.05, Wilcoxon rank sum, for all cell types, **Extended Data Fig 7a**). The motif accessibility changes associated with recovery profiles recapitulated the trend in gene expression and chromatin

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accessibility data, which showed lower accessibility in regions of the genome with closed chromatin, and higher expression in regions of more accessibility (**Extended Data Fig 7b**). In addition, we showed the top 5 up- and down-regulated TFBS motifs per cell type in FP1, RP1, and RP2, which revealed both common and distinct motifs and their accessibility across cell types (**Extended Data Fig 7c-e**). These data also provided further evidence of cell-type specificity in the differences of chromatin and TF accessibility dynamics.

Intra-individual spaceflight responses

To understand individual variation during spaceflight, the coefficient of variation (CV) was measured across microbial, proteomic, cytokine, and gene expression normalized count data, calculated by time interval (pre-flight, in-flight, post-flight, and recovery). Normalized CV scores (see methods) were calculated for each body area from the microbial swabs from both metagenomic and metatranscriptomic data. The oral and forearm microbial variation (**Extended Data Fig 8a-b**) showed *Rothia mucilaginosa* and *Staphylococcus epidermidis* as leading variable strains, but each body site has distinct higher CV species (other body sites are in the **Extended Data Fig 8c-j, Supplementary Table 7**). Similarly, abundance standardized coefficient of variation scores were calculated for EVP proteomic , plasma proteomic, metabolomic, RNA-seq, direct RNA-seq, and cytokine abundance normalized protein and gene counts (**Extended Data Fig 9a-f**). To characterize the subject-to-subject variation and null distributions, we also calculated the differentials for FP1 along with label permutation testing on these calculations to confirm that post-flight CV was not higher than other time points (see methods and **Supplementary Table 8**).

We then utilized the data across missions (I4, Twins, JAXA), assays (scRNA-seq, cfRNA, bulk RNA-seq, proteomics), and cell types (CD4, 8, 14, 16) to find the most recurrent pathways associated with spaceflight. The most enriched pathways were antigen binding, hemoglobin, cytokine signaling, and immune activation,which confirmed the DEG-related and also validated the signatures from our I4 mission **(Fig 5)**. Moreover, the NASA Twins Study data and cell-free RNA profiles from the JAXA study confirmed many of the top-ranked pathways, while also showing differences between bulk blood RNA and protein markers and purified cell populations. These data indicate that leveraging both bulk and sorted cell populations can help clarify signals coming from crew blood dynamics related to spaceflight.

Access to datasets and crew samples

This study introduces full accessibility of astronaut data to the scientific community. Datasets are accessible through an online web portal and scientific data repositories, as well as controlled access to more sensitive data (e.g. genetic variants) and physical specimens. The online web portal splits the data into three data browsers: the SOMA browser, the I4 single-cell browser, and the

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Microbiome Browser (**Extended Data Fig 10a-c**). The SOMA browser (https://soma.weill.cornell.edu/apps/SOMA_Browser/) enables visualization of gene expression (bulk RNA-seq, snRNA-seq, cfRNA-seq, and spatially resolved transcriptomics), mass spectrometry (plasma proteomics, plasma metabolomics, and plasma EVP proteomics), and microbial (metagenomic and metatranscriptomic) data. Gene expression and protein abundance are visualized for each astronaut by timepoint, with fold-change values, statistical significance (q-value), and summary tables. Any selected gene is also then plotted across other missions' data, including the JAXA CFE study, the NASA Twins Study, murine spaceflight datasets from NASA OSDR and Genelab, and controls cohorts. For additional granularity, the single-cell browser (https://soma.weill.cornell.edu/apps/I4_Multiome/) provides visualizations specific to single-cell gene expression (scRNA-seq) and chromatin accessibility (scATAC-seq) data, and includes quality metrics, cell-type annotations, gene co-expression, and chromatin accessibility magnitude estimates. Finally, the microbiome browser contains metagenomic and metatranscriptomic boxplots (https://soma.weill.cornell.edu/apps/I4_Microbiome/) from each timepoint of the study, spanning eight skin locations, deltoid swabs collected pre-skin biopsy, swabs of the SpaceX Dragon capsule, and stool samples.

The remaining biospecimens from this study have been preserved and cataloged for continued use by the scientific community (https://cambank.weill.cornell.edu/). These samples include venous blood plasma, venous blood serum, viably frozen PBMCs, vacutainer red blood cell pellets, urine (both crude and with nucleic acid preservative), extracted saliva nucleic acids (DNA and RNA), extracted whole blood total RNA, extracted skin swab nucleic acids (DNA and RNA), and extracted stool nucleic acids (DNA and RNA). A subset of these samples will be available for additional assays and hypothesis testing by other groups, and the remainder are allocated for long-term biobanking. These first-ever data and specimen resources for astronauts can help enable larger cohorts for increased statistical power and also for new biomedical technologies that will emerge in the future.

Discussion

Overall, these data represent a comprehensive clinical and multi-omic resource from commercial and non-commercial astronaut cohorts, creating the first-ever aerospace medicine biobank, while providing a platform for private citizens to contribute to future astronaut biomedical studies. Additionally, we have demonstrated that short-duration, high-elevation (590km) spaceflight results in broad-ranging molecular changes, wherein some of these changes mirror what has been observed during longer-duration spaceflight, including elevated cytokines, telomere elongation, and gene expression changes for immune activation, DNA damage response, and oxidative stress. While >95% of markers return to baseline in the months after the mission has ended, some proteins, genes, and cytokines appear to be activated only in the recovery period after spaceflight and persist post-flight for at least 3 months.

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These results collectively indicate a dynamic recovery profile that substantially reverses the direction of differential gene expression in multiple key biological pathways from the post-flight timepoint (R+1) and afterward. This suggests that re-adaptation to Earth activates a range of restorative mechanisms that help recover, at least in part, the physiological stress imposed by exposure to the space environment. The systematic analysis of the molecular and cellular changes observed post-flight afford us with a unique opportunity to capture naturally occuring health restoring mechanisms, which can be used for therapeutic target discovery. Furthermore, we observe a nuanced regulatory landscape, where enriched recovery pathways are both unique to individual cell types and span cell types in unique combinations. We have carefully indexed a variety of profiles for flight (FP), recovery (RP), and longitudinal (LP) analysis that are annotated in processed data files available in NASA OSDR. The RP1 profile indicates we need more frequent sampling in the 1-2 months directly after flight to untangle the gene and pathway responses during readaptation from spaceflight. This will be especially true during longer missions, in which readaptation will likely be more intense.

Of note, these data are the first-ever joint single nuclei chromatin profiles (RNA and ATAC) for astronauts, and they also leverage new methods that can track gene expression and epigenetic changes within the same cells. This single-cell, dual-measurement assay provides new data on the molecular changes and regulatory response to spaceflight (e.g. chromatin and TFBS accessibility), and the data revealed distinct levels of stress and adaptation by each cell type. Specifically, the monocyte cell populations (CD14, CD16) had, by far, the largest change in expression and response than any other cell type (e.g. T-cells, B-cells, NK cells). The differences between PBMC sub-populations also suggests that single-cell sequencing can be helpful for delineating unique cell type responses in future studies as well. Indeed, while the immune system and hematopoietic systems both show thousands of transient changes at the gene expression level, the chromatin architecture is distinctly disrupted, both in scale and duration, in these CD14+ and CD16+ monocyte populations. These cell types were also found to be disrupted in the NASA Twins Study[7,23], and thus represent a key cell type to be studied for future missions.

The cfRNA, direct RNA, and spatial RNA profiles revealed unique profiles that differed based on recovery and longitudinal analysis, suggesting that the multi-omic footprint of spaceflight is much wider than previously observed. While some focused studies of cfRNA have shown that it can be used to detect mitochondrial increase in blood related to spaceflight[7–9,24], it had never been applied as a “full body molecular scan” to detect differential tissue and cell stress until now. Notably, the SOMA resource enables comparisons of the cell-free and cell-encapsulated RNAs, as well as the exosomal fractions, all within the same framework, which is essential for delineating and ranking the cells and tissues in the body that are the most disrupted by spaceflight. Similar to the utility of single-cell data[23], the granularity of seeing cell lysis from all tissues across the body in one assay makes cfRNA profiling an ideal addition to some of the

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proposed standard measures for spaceflight monitoring. So far, cfRNA data indicate that each part of the body may show its own transcriptional response, shedding rate, and/or RNA excretion rates, and thus each tissue should be examined on its own and then compared to other sites and assays. Similarly, in the microbiome data, the taxonomic classifications yielded differences in variation by body site, time point, and crew member (**Extended Data Fig. 8**), indicating the need for multi-site and multi-omic sampling for ideal understanding of microbiome changes associated with spaceflight.

However, this study is not without limitations. Although cell-specific comparisons could be made between the I4 mission and prior studies, the assays and collection protocols were slightly different (e.g. column purification vs. droplet-sorted cells), and thus comparisons will be imperfect. Also, while the same tube types and methods were employed whenever possible, cross-mission comparisons will inevitably include noise from various other types of technical variation, including batches of library preparation and extraction kits, slightly different collection intervals (L-3 vs. L-10), and/or different sequencing/profiling technologies (e.g. Illumina vs. Ultima or LC-MS/MS vs. NULISASeq). Finally, since our comparisons to other NASA and JAXA datasets span, at most, 46 astronauts, these data are not sufficient to guide medical interventions or inference of mechanisms. As such, these SOMA data and resources should be viewed as preliminary molecular maps of the human body's response to spaceflight.

Future Directions

Nonetheless, the data from the I4 mission will be an invaluable resource for future studies. Deep dives into secretome profiling[25], single-nuclei multiome[26], viral activation and ecological restructuring[27], skin spatial transcriptomic profiling[28], epitranscriptomic profiling[22], and genome integrity and clonal hematopoiesis[29] have already leveraged this resource. In future missions, additional biomedical profiling can help delineate the short and long-term health impacts of spaceflight, including changes in telomere length dynamics, DNA methylation, ncRNAs, as well as additional sample types, such as hair follicles, tears, sperm, and other biospecimens. Indeed, the remaining samples from the I4 mission have been biobanked for just this reason, and to help the scientific community tackle future objectives[12]. Additionally, aliquots of DNA, RNA, protein, serum, urine, and stool from the protocols performed in this paper have been consented for release and are available (https://cambank.weill.cornell.edu/) for request , whereby researchers can then append new results to this extensible SOMA repository.

Differences between crew members' biomedical and cellular responses may be caused by several factors, including inter-individual genetic differences (**Figure 5**), duration of the mission, the higher flight elevation (590km), the unique environment of the SpaceX Dragon capsule, or a combination of these and other factors. To address these hypotheses, other molecular assays can be informative for future missions, including other epigenetics and chromatin conformation

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assays. For example, we have investigated using the method Cleavage Under Targets and Release Using Nuclease (Cut&Run) to profile histone modifications using 5,000 and 10,000 T cells as assay input (**SI Fig 3a**), down from the 10 million cells recommended in the original protocol[30]. T cells were collected from C002 during the R+194 recovery time point, but show high variance at lower input. Nonetheless, the combination of current modifications profiled can be used to annotate active enhancer regions in the genome (**SI Fig 3b**), which is a novel profile for astronauts, and opens the door to better understanding of the gene regulatory changes induced by spaceflight for each crew member.

Molecular changes described here can also help guide research and countermeasure development, but are only the start of the process of mitigating risks, especially since year-long and multi-year space habitat missions will represent greater biomedical challenges. To aid in this effort, the SOMA resource will continue to expand as further samples are sequenced from the I4 mission[11,12] and samples are collected from future missions that travel farther into space, and then compared to other longitudinal, multi-omics cohorts[31]. Biospecimen samples have been collected and processed from Polaris and Axiom crews, and this Atlas also represents an open call for research participation for astronauts from any commercial or governmental programs.

Finally, while the multi-omic data and resources from the I4 crew represent the largest release of astronauts' data to date, the I4 crew is still a small cohort, and represents only the first step towards resolving the many hazards and needs for long-term missions[32], building towards enough statistical power, and contextualization of normal human biological variation[33,34]. Fortunately, one of crew members for I4 will continue to donate to the SOMA Biobank for multiple missions, including the Polaris Dawn mission, and data and samples from additional crews (e.g. long-term Twins Study follow-up, Axiom-2/Saudi Arabia, JAXA, and Malta) are now being collected and integrated as well. These cross-mission datasets create a unique opportunity for long-term, in-depth analysis of the impact of spaceflight on the human body. Such data are especially important as missions travel farther away from Earth[35], and for longer periods of time, where the data and biomedical discoveries can help prepare commercial and state-sponsored agencies for the lunar, Martian, and exploration-class missions.

References

1. Bushnell, D. M. & Moses, R. W. *Commercial Space In The Age Of 'New Space', Reusable Rockets and The Ongoing Tech Revolutions*. https://ntrs.nasa.gov/citations/20180008444 (2018).
2. Lang, T. *et al.* Towards human exploration of space: the THESEUS review series on muscle and bone research priorities. *NPJ Microgravity* **3**, 8 (2017).

ACCELERATED ARTICLE PREVIEW

3. Martin Paez, Y., Mudie, L. I. & Subramanian, P. S. Spaceflight Associated Neuro-Ocular Syndrome (SANS): A Systematic Review and Future Directions. *Eye Brain* **12**, 105–117 (2020).
4. Crucian, B. E. *et al.* Immune System Dysregulation During Spaceflight: Potential Countermeasures for Deep Space Exploration Missions. *Front. Immunol.* **9**, 1437 (2018).
5. Trudel, G., Shahin, N., Ramsay, T., Laneuville, O. & Louati, H. Hemolysis contributes to anemia during long-duration space flight. *Nat. Med.* **28**, 59–62 (2022).
6. Charvat, J. M. *et al.* Long-term Cardiovascular Risk in Astronauts: Comparing NASA Mission Astronauts With a Healthy Cohort From the Cooper Center Longitudinal Study. *Mayo Clin. Proc.* **97**, 1237–1246 (2022).
7. Garrett-Bakelman, F. E. *et al.* The NASA Twins Study: A multidimensional analysis of a year-long human spaceflight. *Science* **364**, (2019).
8. Muratani, M. Cell-free RNA analysis of plasma samples collected from six astronauts in JAXA Cell-Free Epigenome (CFE) Study. NASA GeneLab https://doi.org/10.26030/R2XR-H714 (2022).
9. Rutter, L. Extracellular mitochondria associated with scavenger receptor CD36 as a hallmark of stress response in space. *Nat. Commun.*
10. Schmidt, M. A., Schmidt, C. M., Hubbard, R. M. & Mason, C. E. Why Personalized Medicine Is the Frontier of Medicine and Performance for Humans in Space. *New Space* **8**, 63–76 (2020).
11. Christopher Jones, ..., Christopher E. Mason. The SpaceX Inspiration4 mission reveals inflight molecular and physiological metrics from an all-civilian crew. *Nature*.
12. Eliah G. Overbey, ..., Christopher E. Mason. Collection of Biospecimens from the

ACCELERATED ARTICLE PREVIEW

Inspiration4 Mission Establishes the Standards for the Space Omics and Medical Atlas (SOMA). *Nat. Commun.*

13. Downey, P. & Peakman, T. C. Design and implementation of a high-throughput biological sample processing facility using modern manufacturing principles. *Int. J. Epidemiol.* **37 Suppl 1**, i46–50 (2008).
14. Kulu, E. In-space economy in 2021—statistical overview and classification of commercial entities. in *72nd International Astronautical Congress (IAC 2021), Dubai, United Arab Emirates* 25–29 (2021).
15. Berrios, D. C., Galazka, J., Grigorev, K., Gebre, S. & Costes, S. V. NASA GeneLab: interfaces for the exploration of space omics data. *Nucleic Acids Res.* **49**, D1515–D1522 (2021).
16. Scott, R. T. *et al.* Advancing the Integration of Biosciences Data Sharing to Further Enable Space Exploration. *Cell Rep.* **33**, 108441 (2020).
17. Scott, R. T. *et al.* Biomonitoring and precision health in deep space supported by artificial intelligence. *Nature Machine Intelligence* **5**, 196–207 (2023).
18. Wu, F. Single Cell Analysis Identifies Conserved Features of Immune Dysfunction in Simulated Microgravity and Spaceflight. *Nat. Commun.*
19. Luxton, J. J. *et al.* Telomere Length Dynamics and DNA Damage Responses Associated with Long-Duration Spaceflight. *Cell Rep.* **33**, 108457 (2020).
20. Luxton, J. J. *et al.* Temporal Telomere and DNA Damage Responses in the Space Radiation Environment. *Cell Rep.* **33**, 108435 (2020).
21. Loy, C. J. *et al.* Nucleic acid biomarkers of immune response and cell and tissue damage in children with COVID-19 and MIS-C. *Cell Rep Med* **4**, 101034 (2023).

ACCELERATED ARTICLE PREVIEW

22. Grigorev, K. Direct RNA sequencing of astronauts reveals spaceflight-associated epitranscriptome changes and stress-related transcriptional responses. *Nat. Commun.*
23. Gertz, M. L. *et al.* Multi-omic, Single-Cell, and Biochemical Profiles of Astronauts Guide Pharmacological Strategies for Returning to Gravity. *Cell Rep.* **33**, 108429 (2020).
24. Bezdan, D. *et al.* Cell-free DNA (cfDNA) and Exosome Profiling from a Year-Long Human Spaceflight Reveals Circulating Biomarkers. *iScience* **23**, 101844 (2020).
25. Nadia Houerbi, ..., Christopher E. Mason. Secretome profiling captures acute changes in oxidative stress, brain homeostasis and coagulation from spaceflight. *Communications Biology*.
26. JangKeun Kim, Braden Tierney, ..., Christopher E. Mason. Single-cell multi-ome and immune profiles of the Inspiration4 crew reveal conserved, cell-type, and sex-specific responses to spaceflight. *Nature*.
27. Braden Tierney, ..., Christopher E. Mason. Viral activation and ecological restructuring characterize a microbiome axis of spaceflight-associated immune activation.
28. Park, J. Spatial multi-omics of human skin reveals KRAS and inflammatory responses to spaceflight. *Nat. Commun.*
29. J. Sebastian Garcia Medina, ..., Christopher E. Mason. Genome and Clonal Hematopoiesis Stability Contrasts with Immune, cfDNA, mitochondrial, and telomere length changes to Short Duration Spaceflight.
30. Skene, P. J. & Henikoff, S. An efficient targeted nuclease strategy for high-resolution mapping of DNA binding sites. *Elife* **6**, (2017).
31. Zhou, W. *et al.* Longitudinal multi-omics of host–microbe dynamics in prediabetes. *Nature* **569**, 663–671 (2019).

ACCELERATED ARTICLE PREVIEW

32. Afshinnekoo, E. *et al.* Fundamental Biological Features of Spaceflight: Advancing the Field to Enable Deep-Space Exploration. *Cell* **183**, 1162–1184 (2020).

33. Song, Z. *et al.* Lifestyle impacts on the aging-associated expression of biomarkers of DNA damage and telomere dysfunction in human blood. *Aging Cell* **9**, 607–615 (2010).

34. Barnett, M., Young, W., Cooney, J. & Roy, N. Metabolomics and proteomics, and what to do with all these 'omes': Insights from nutrigenomic investigations in New Zealand. *J. Nutrigenet. Nutrigenomics* **7**, 274–282 (2014).

35. Nangle, S. N. *et al.* The case for biotech on Mars. *Nat. Biotechnol.* **38**, 401–407 (2020).

Figure Legends

Figure 1: Compendium of Astronaut Omic Data and Time-Series Analysis Paradigms (a) Omics and biochemical assays were performed on blood (whole blood, serum, PBMCs, plasma, plasma-derived EVPs, dried blood spots), oral (microbiome swabs), nasal (microbiome swabs), skin (biopsy, microbiome swabs), environmental (env.) (microbiome swabs), and excrement (excrem.) (urine, stool) samples. **(b)** The timepoints of this study are separated into four different categories: pre-flight (L-92, L-44, L-3), in-flight (flight day (FD): FD1, FD2, FD3), post-flight (R+1), and recovery (R+45, R+82, R+194). Colored circles indicate which assay was performed at each timepoint. Assays were performed on all crew members, unless denoted with an asterisk '*'. **(c)** Indicator for which assay types have been previously performed in spaceflight studies, broken down by the NASA Twins Study, JAXA studies, and anonymized NASA cohort studies.

Figure 2: Telomere and Cytokine Twins Study Comparison. (a) Normalized average telomere lengths for i4 crewmembers, pre-flight, during flight, post-flight, determined by qPCR analyses of blood (DNA) collected on DBS cards (n=32 samples for 4 independent subjects across 8 timepoints). Two-sided p-values were derived using a mixed effects linear model that incorporated fixed effects for different timepoints (Pre-Flight, In-Flight, Post-Flight, Recovery) and random effects to account for variations among subjects. **(b)** Changes in down-regulated (purple) and up-regulated (orange) gene expression $\log_2$ fold-change directionality post-flight from the Twins Study vs Inspiration4 in CD19 B cells , CD4+ T cells, and CD8+ T cells (statistical significance by two-sided Wilcoxon rank-sum test). Number of genes are shown below the violin plots. **(c)** Relative cytokine/chemokine abundance pre-flight, post-flight, and during recovery in the Inspiration4 crew vs the NASA Twins Study and anonymized NASA astronaut cohorts for CCL2, IL10, and IL6. (MLBT: Multiplexing LASER Bead Technology). **(d)** Relative

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cytokine/chemokine abundance of BDNF and IL19 pre-flight, post-flight, and during recovery in the Inspiration4 crew. In **a** the center of the boxplots represent the median, the box hinges encompass the first and third quartiles, and whiskers extend to the smallest and largest value no further than the 1.5x interquartile range (IQR) away from the hinges. In **b**, the center white dot represents the median, and the white line shows the range of the first and third quartiles. In **c, d** the two-sided p-values and adjusted q-values were derived using a mixed effects model that incorporated fixed effects for different timepoints (Pre-Flight, In-Flight, Post-Flight, Recovery) and random effects to account for variations among subjects, except in the Twins Study which had a single subject (n=1). P-values with an asterisk have a q-value>0.05 after multiple correction testing.

Figure 3: Body-wide Tissue Stress Map with Cell-Free RNA. (a) Cell type deconvolution using Bayes Prism with Tabula Sapiens as a reference. Top 10 cell types by average fraction across all samples with all remaining cell types summed together as 'other'. **(b)** Cell type of origin for hepatocytes, endothelial cells, hematopoietic stem cells, and melanocytes, which all show increased abundance during post-flight and recovery timepoints. **(c)** Cell proportion changes in different layers of the skin from spatially resolved transcriptomics on skin biopsies. Predicted melanocyte abundance changes are significant in the inner epidermal and outer dermal skin compartments. In **b-c** n=4 independent subjects across 7 timepoints, and the center of the boxplots represent the median, the box hinges encompass the first and third quartiles, and whiskers extend to the smallest and largest value no further than the 1.5x interquartile range (IQR) away from the hinges.

Figure 4: Recovery Profile Dynamics in PBMCs. (a) Number of DEGs from PBMC snRNA-seq for each cell type during the flight, recovery, and longitudinal profiles (adj two-sided p-value < 0.05, logFC $> |0.5|$). **(b)** Fraction of DEGs shared with FP1 at RP1, RP2, and LP2 for each cell type. **(c)** Directionality of logFC between flight profile FP1 and recovery profiles RP1 and RP2 for DEGs present in both profiles. **(d)** Barchart of pathways of DEGs present in RP1 that were absent in FP1 in monocytes and T cells. Bars are shaded by FDR value and the enrichment ratio is on the x-axis.

Figure 5: Pathway Enrichment of Most Variable Genes. Enriched pathways in postflight compared to preflight across various assays and missions, analyzed using fGSEA. Color represents the normalized enrichment score, while dot size indicates Benjamini-Hochberg adjusted q-values. Only the pathways with unadjusted p-value < 0.01 are shown. Bar plot shows the total number of comparisons with $q < 0.05$ for every pathway, colored by the direction of the enrichment. The column "Mixed cell type" refers to whole blood for I4 data and lymphocyte depleted cells for the NASA Twins Study.

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Methods

IRB Statement

All subjects were consented at an informed consent briefing (ICB) at SpaceX (Hawthorne, CA), and samples were collected and processed under the approval of the Institutional Review Board (IRB) at Weill Cornell Medicine, under Protocol 21-05023569. All crew members have consented for data and sample sharing.

I4 Data Compendium Data Generation

Full methodology for sample preparation, nucleic acid/protein extraction, sequencing, mass spectrometry, and analysis are reported in Supplementary Note 1. Sample collection is reported in (Overbey *et al.*)[12].

I4 vs NASA Twins Study Cytokine Analysis

Differential abundance analyses of cytokines and other analytes were conducted using a mixed-effects model. This model included fixed effects for time, treated as a categorical variable with levels corresponding to preflight (L-92, L-44, L-3), immediate return (R+1), and recovery periods (R+45, R+82, R+194). Additionally, subject-specific effects were incorporated as random effects in the model. The p-values for the coefficients obtained from this model were adjusted for multiple comparisons using the Benjamini-Hochberg procedure.

10X Genomics snRNA-seq Negative Control Data

We acquired cryovials of PBMCs from a healthy male aged 22 (AllCells) and stored them in vapor phase liquid nitrogen cryotanks. Two cryovials from this donor were thawed and processed on two different days in the same week, at the same lab, using the 10x Genomics Demonstrated Protocol called Nuclei Isolation for Single Cell Multiome ATAC + Gene Expression Sequencing (CG000365). From each day's nuclei suspension, ATAC and Gene Expression libraries were generated (in technical triplicates on the same chip) according to the Chromium Next GEM Single Cell Multiome ATAC + Gene Expression User Guide (CG000338) and sequenced on a NovaSeq 6000 Sequencer. The resulting single-nuclei GEX files were aligned using the cellranger-arc pipeline (v6.1.1) from 10X genomics against human reference genome hg38. Quality control and cell annotation was performed on the snGEX gene-cell matrices using the R Seurat package (version 4.2.0)[36]. Subpopulations were clustered and labeled using a publicly available Azimuth human PBMCs reference[37] in conjunction with Seurat's supervised clustering functionality. Differential gene expression analysis for labeled subpopulations was performed using the FindMarkers functionality of Seurat, with a log fold change (logFC) cutoff point of 0.5. P-values of resultant genes were measured using the Wilcoxon Rank Sum test and deemed significant for $p < 0.05$. Correlation of chromatin accessibility change and gene expression change in FP1 was characterized by summing up the accessibility in a promoter window consisting of the transcription start site (TSS) ±500bp for

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every cell in a given cell type across every gene, normalizing it to counts per million, taking the $\log_2$ difference between the different timepoints, and matching it to the $\log_2$ fold-change from the differential RNA analysis.

Recovery Profile Analysis

Differentially expressed genes from each cell type from the PBMC 10X Genomics single-cell data were filtered for logFC > |0.5| and adj p-val < 0.05 for the FP1, RP1, RP2, and LP2 profiles. UpSet plot of shared genes was generated using Intervene[38]. Over-represented pathways were calculated using WebGestalt [39] using the default parameters and the KEGG database. Pathways included in the analysis were within the top 40 lowest FDR values or all pathways with an FDR under 0.05.

TF motif accessibility analysis

chromVAR[40] was used for the analysis of sparse chromatin accessibility from 10X Genomics single-cell data. The FindMarkers function of the Seurat package was used for differential analysis with setting mean.fxn as rowMeans and fc.name = "avg_diff". Top 5 differentially accessible TF motifs from each cell type from the PBMC 10X Genomics single-cell data for the FP1, RP1, and RP2. Heatmaps were generated by the ComplexHeatmap R package.

Calculation of Individual Variation

In order to distinguish the analytes that varied with the greatest magnitude between individual crew members (C001, C002, C003, C004), a coefficient of variation (CV) calculation was applied across measurements collected for each: metagenomic species, metatranscriptomic species, plasma protein abundance, EVP protein abundance, metabolite abundance, cytokine abundances (Alamar Bio), and gene expression values (Oxford Nanopore dRNA-seq, Ultima RNA-seq). CV calculations were performed using the formula 'np.std(x, ddof=0) / np.mean(x)', where 'x' is the array of normalized analyte values and 'np' refers to the numpy scientific computing package (v=1.26). CV calculations were split into "Pre-Flight" (L-92, L-44, L-3), "Post-Flight" (R+1), and "Recovery" (R+45, R+82) intervals. The metagenomic and metatranscriptomic data also includes the "In-Flight" (FD2, FD3) interval.

Before performing the CV calculation, data was first normalized. The "limma" R package was applied to the plasma proteomic, EVP proteomic, and metabolite data (version 3.52). DESeq2 (1.36.0) was applied to the Oxford Nanopore dRNA-seq and Ultima RNA-seq data. For microbiome data, to measure individual contributions per astronaut and for each tissue type ("Armpit", "Forearm", "Nasal", "Oral", "Post-Auricular", "T-Zone", etc), we averaged MetaPhlAn4, species-level, relative abundance values across our collected sample points. CVs were calculated for individual genes, proteins, and microbial organisms.

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An abundance standardized CV calculation was also performed to correct for the mean/CV relationship in the assays. First, the normalized abundance values across all the analytes of a given dataset were split into approximately 100 mean-abundance quantiles (*M_q* where *q* is 1 to 100 or fewer, in case it resulted in less than 10 points per quantile group). For each quantile group, the mean and the standard deviation of CV values for the analytes belonging to *M_q* were calculated (*CVmean_q* and *CVstd_q*) to prepare the mean/CV distribution reference set across all timepoints and samples. Then, for a given timepoint, the *CV* calculated for each analyte was matched to the corresponding *CVmean_q* and *CVstd_q* based on cross-referencing to the *M_q* interval overlap. Finally the abundance standardized CV was calculated as the z-score of the CV as *(CV - CVmean_q)/(CVstd_q)*. Permutation testing of the Postflight versus Preflight difference was performed by repeating this procedure 10,000 times on shuffled labels for samples, taking the difference Rank and z-score of the observed value was calculated against the random permutations to order the genes and used as input to fgsea for pathway enrichment analysis. Gene Ontology analysis on abundance standardized CVs was performed with Enrichr[41,42] using the default settings.

Data Availability

Datasets have been uploaded to two data repositories: the NASA Open Science Data Repository (osdr.nasa.gov; comprised of NASA GeneLab[15] and the NASA Ames Life Sciences Data Archive [ALSDA][16,32]). Identifiers for publicly downloadable datasets in the OSDR are documented in Supplementary Table 9. Data can be visualized online through the SOMA Data Browser (https://soma.weill.cornell.edu/apps/SOMA_Browser), the single-cell expression and chromatin browser (https://soma.weill.cornell.edu/apps/I4_Multiome/), and the microbiome browser (https://soma.weill.cornell.edu/apps/I4_Microbiome/).

Code Availability

Code for data processing is available at https://github.com/eliah-o/inspiration4-omics.

Methods References

36. Hao, Y. *et al.* Integrated analysis of multimodal single-cell data. *Cell* **184**, 3573–3587.e29 (2021).

37. Stuart, T. *et al.* Comprehensive Integration of Single-Cell Data. *Cell* **177**, 1888–1902.e21 (2019).

38. Khan, A. & Mathelier, A. Intervene: a tool for intersection and visualization of multiple gene or genomic region sets. *BMC Bioinformatics* **18**, 287 (2017).

39. Wang, J., Vasaikar, S., Shi, Z., Greer, M. & Zhang, B. WebGestalt 2017: a more

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comprehensive, powerful, flexible and interactive gene set enrichment analysis toolkit. *Nucleic Acids Res.* **45**, W130–W137 (2017).

40. Schep, A. N., Wu, B., Buenrostro, J. D. & Greenleaf, W. J. chromVAR: inferring transcription-factor-associated accessibility from single-cell epigenomic data. *Nat. Methods* **14**, 975–978 (2017).

41. Chen, E. Y. *et al.* Enrichr: interactive and collaborative HTML5 gene list enrichment analysis tool. *BMC Bioinformatics* **14**, 128 (2013).

42. Kuleshov, M. V. *et al.* Enrichr: a comprehensive gene set enrichment analysis web server 2016 update. *Nucleic Acids Res.* **44**, W90–7 (2016).

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Competing Interest

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BTT is compensated for consulting with Seed Health and Enzymetrics Biosciences on microbiome study design and holds an ownership stake in the former. KLB receives research funding from Servier and Bristol Myers Squibb and serves on the medical advisory board of GoodCell. SC Employee and shareholder at NanoString Technologies. KB, JM, AB, JZ, BL, AA, SK, SL are employees of and have a financial interest in Element Biosciences. EEA is a consultant for Thorne HealthTech. ASG, LW, PT, QY, JC, RB, AS, and DH are employees of and have a financial interest in Seer Inc. and Prognomiq Inc. CM is compensated by Thorne HealthTech. CEM is co-Founder of Cosmica Biosciences. DCL and IM receive research grant support/funding from Atossa Inc. RDG is on the scientific advisory board of Elysium Health and is an advisor to Gore Range Capital. RDG is also an informal advisor to BelleTorus Corporation but has no financial ties to BelleTorus at this time. Caleb M. Schmidt, Julian C. Schmidt, and Michael A. Schmidt hold shares in Sovaris Holdings, LLC. Jan Krumsiek holds equity in Chymia LLC and IP in PsyProtix and is cofounder of iollo. Min Yu is the founder and president of CanTraCer Biosciences Inc. GC COI list: arep.med.harvard.edu/gmc/tech.html. AM has research funding from Jannsen, Epizyme, and Daiichi Sankyo and has consulted for Treeline, AstraZeneca and Epizyme. Authors not listed here do not have competing interests.

Additional Methods and Details

In-depth methods and protocol information for all assays in **Fig 1A** is located in Supplementary Note 1.

Extended Data Figure Legends

Extended Data Figure 1: Prior Work and Comparative Profiles. (a) Prior human spaceflight omics study and sample counts with publicly available data, which is housed in OSDR. **(b)** Total number of sequence nucleic acid molecules in all prior studies (OSDR, blue) compared to this study (Inspiration4, red). **(c)** Visualization of the different analysis paradigms used when analyzing the time-series spaceflight data. The database identifier is provided as a shorthand to reference each comparison.

Extended Data Figure 2: Pipelines Overview. Computational pipelines for **(a)** 10X Genomics Multiome sequencing (snRNA and snATAC), **(b)** ONT direct RNA-sequencing gene expression and m6A detection, **(c)** Nanostring GeoMx whole transcriptome atlas profiling, **(d)** cfRNA gene abundances, **(e)** T-cell repertoire and B-cell repertoire V(D)J immune profiling, **(f)** plasma processing for proteomic, metabolomic, and EVP proteomic profiling, and **(g)** microbial profiling.

Extended Data Figure 3: Transcriptomic Fingerprint of Short Duration Spaceflight. (a) Representative spatial imaging of skin biopsy tissue for each crew member (C001, C002, C003, and C004). Processing was done in two batches and across four ROI types. **(b)** UMAP projection of the ROIs. The colors represent ROI types and shapes represent time points. Most of the ROIs

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showed good clustering around ROI types in both time points. **(c)** Heatmap visualization of top variable genes across ROIs and time points in skin biopsies. **(d)** PCA on scaled vst normalized counts of top 500 variable genes in cfRNA data. **(e)** Z-score of vst normalized cfRNA abundances from DESeq2 (BH adjusted two-sided p-value < 0.01, |log2FC| > 1). Total of 927 genes. **(f)** cfDNA RNA species elevated pre-flight vs post-flight. Top 500 displayed for each group ranked by $\log_2$ fold-change.

Extended Data Figure 4: Direct RNA-seq Gene Expression and RNA m^{6}A Modifications Across 13 Comparative Profiles. (a) Patterns of gene expression across seven time points and 13 comparisons. *Left:* z-scored log-transformed normalized gene counts obtained from salmon (bottom left of each cell) and featureCounts (top right of each cell). *Right:* $\log_2$(fold-change) values obtained from edgeR (bottom left of each cell) and from DESeq2 (top right of each cell). The genes are clustered by z-scored log-transformed normalized counts using the correlation distance metric. **(b)** Patterns of base-level m^{6}A modifications across seven time points and 13 comparisons. *Left:* z-scored log-transformed positional methylation probabilities obtained from m6anet. *Right:* percentage of change in methylation between the conditions in each comparison, obtained from methylKit. The sites are clustered by the pattern of differential methylation across all comparisons using the correlation distance metric. On both panes **(a)** and **(b)**, only the genes and sites with significant differences in expression and/or methylation in at least one comparison are plotted; the significance of individual comparisons is annotated with up and down arrows.

Extended Data Figure 5: Single-Nuclei RNA-Sequencing Controls. (a) The number of DEGs in I4 flight profiles (FP1: grouped crew or individual crew members), I4 longitudinal profiles (LP3: grouped crew or individual crew members), and negative control groups (mock control day 2 vs day 1, mock control same day technical replicate differences, and I4 inter-subject comparisons in preflight). All comparisons were done with downsampling to the same number of cells. Intra-timepoint subject comparisons are crew-to-crew comparison in the pre-flight (L-92, L-44, L-3), immediately post-flight (R+1), and recovery (R+45, R+82) time intervals. The bars show the mean of the total number of DEGs, and error bars show the standard error for groups that have 3 or more comparisons summarized. (n=4 independent subjects with 6 timepoints, and n=1 control subject with 2 timepoints and 3 technical scRNAseq replicates for each timepoint.) **(b)** The number of total DEGs identified by DESeq2 and pseudobulk counts in I4 flight profiles (FP1), I4 longitudinal profiles (LP3), and negative control groups (mock control day 2 vs day 1). All comparisons were done with aggregation into a single sample for each crew and each cell type. The bars show the mean of the total number of DEGs. **(c)** DEG directionality heatmap represents the overlap of up-regulated (orange) and down-regulated (purple) DEGs across comparison groups (I4 timepoint comparisons, I4 individual variation, 10X negative controls). **(d)** Heatmap representing the $\log_2$ fold-change of I4-FP1 up-regulated and down-regulated PBMC DEGs in each comparison group (I4 timepoint comparisons, I4 individual variation, 10X Genomics negative controls).

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Extended Data Figure 6: Recovery Profile Analysis. (a) DEGs shared between the T cell and monocyte lineages for DEGs present in RP1, but not present in the FP1 profile. **(b)** Overrepresented KEGG pathways from DEG sets unique to the RP1 profile in the t-cell and monocyte lineages. The percent of pathways unique to each cell type are quantified along with the various configurations in which the pathway is shared between cell types and lineages.

Extended Data Figure 7: Single-Nuclei ATAC-seq and TFBSs. (a) log2 fold-change of chromatin accessibility at promoters of genes that are differentially expressed in FP across different cell types. Promoters are defined as the transcription start site ± 500 bp of a given gene. Two-sided p-values were calculated by Wilcoxon rank-sum test. Violin plots show the density of the points, and the center white dot represents the median, and the white line shows the range of the first and third quartiles. **(b)** Directionality of delta z-score for motif accessibility between flight profile FP1 and recovery profiles RP1 and RP2 for significant TF motifs of FP1 present in either RP1 or RP2. **(c)** Heatmap of accessibility z-score of top5 significantly increased and decreased TFs in each cell type in FP1. **(d)** Heatmap of accessibility z-score of top5 significantly increased and decreased TFs in each cell type in RP1. **(e)** Heatmap of accessibility z-score of top5 significantly increased and decreased TFs in each cell type in RP2.

Extended Data Figure 8: Coefficients of Variation (CVs) for Microbial Taxa Across Body Sites. (a-j) CVs for different microbial taxa from oral, forearm, nasal, gluteal crease, occiput, axillary vault, umbilicus, glabella, post-auricular, and toe web space regions generated from skin swabs. CVs are calculated from both metagenomic and metatranscriptomic sequencing data.

Extended Data Figure 9: CV Analysis of Datasets by Time Interval. (a-f) Abundance standardized CVs for the human omics assays across pre-flight, post-flight, and recovery time intervals. The most variable analytes are labeled at the top of each violin plot. CV is calculated across n=4 independent subjects in 6 to 7 timepoints for p=16283, 8464, 527, 1765, 656, 203 analytes respectively from a to f. The center dot represents the median, and the black line shows the range of the first and third quartiles.

Extended Data Figure 10: SOMA Web Portals. (a) Three different web portals were created: SOMA Browser, Single-Cell Browser, and Microbial Browser. The SOMA Browser includes **(b)** gene expression and protein abundance measurements in line chart and volcano plot formats, **(c)** log fold change calculations, and **(d)** comparison of DEGs across different contrasts, assays, studies, and organisms (statistical tests dependent on comparison, please see website). **(e)** The Single-Cell Browser enables visualization of cell type specific information, including gene co-expression and ATAC-seq region peak visualization. **(f)** The Microbial Browser includes microbial abundances across timepoints from different annotation databases for metagenomic and metatranscriptomic datasets (n=4 independent subjects for 8 timepoints and 12 collection

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sites, or n=9 environmental samples for 4 timepoints). The center of the boxplots represent the median, the box hinges encompass the first and third quartiles, and whiskers extend to the smallest and largest value no further than the 1.5x interquartile range (IQR) away from the hinges.

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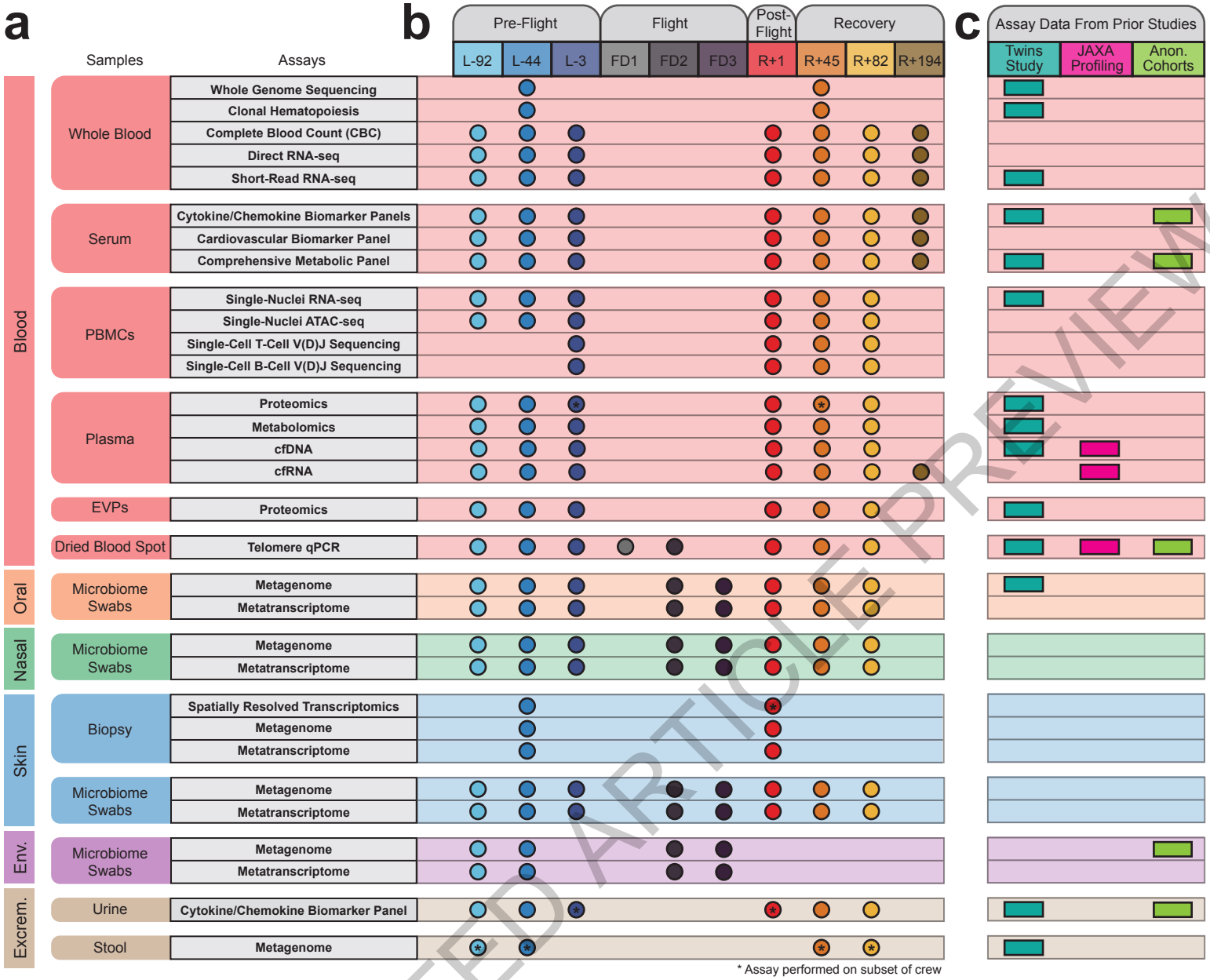

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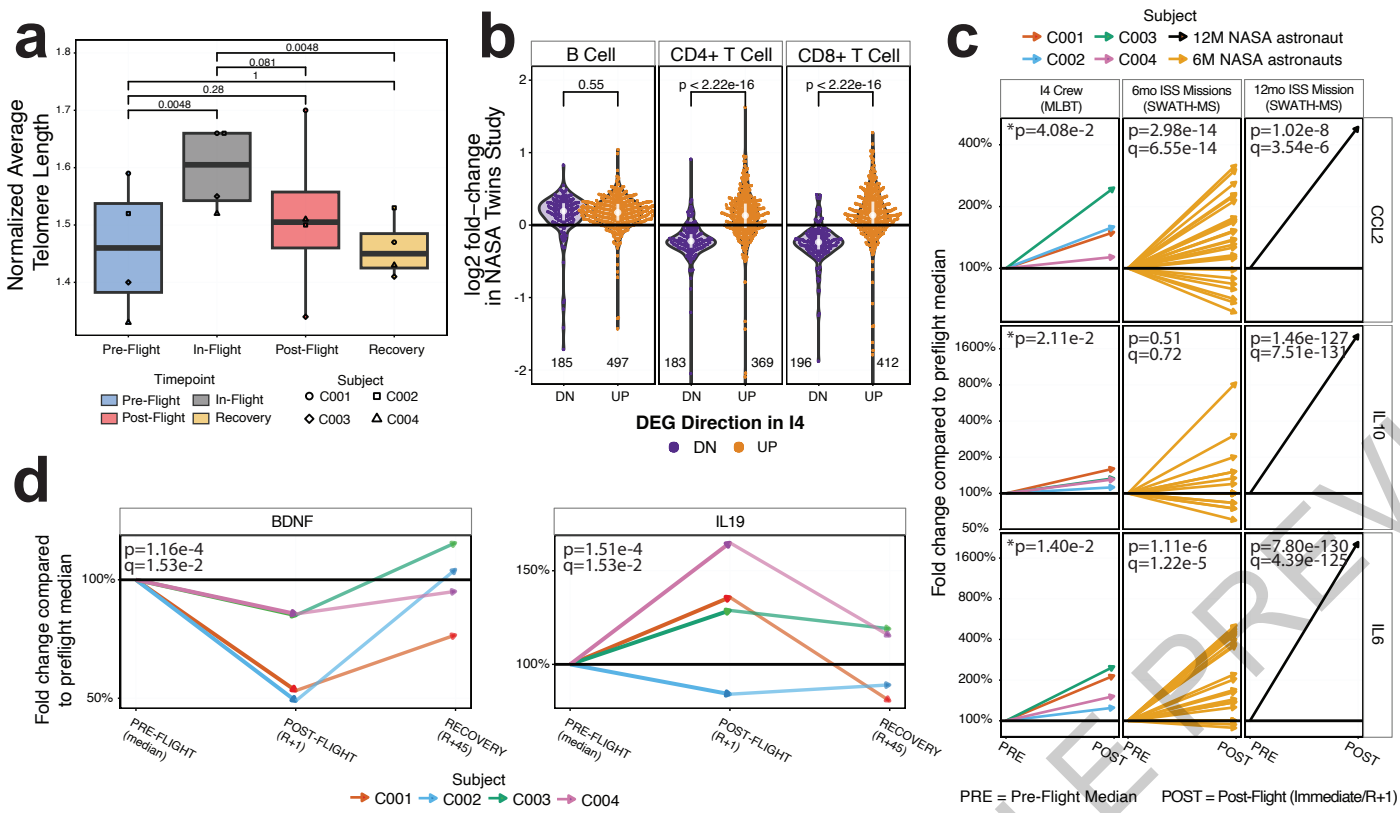

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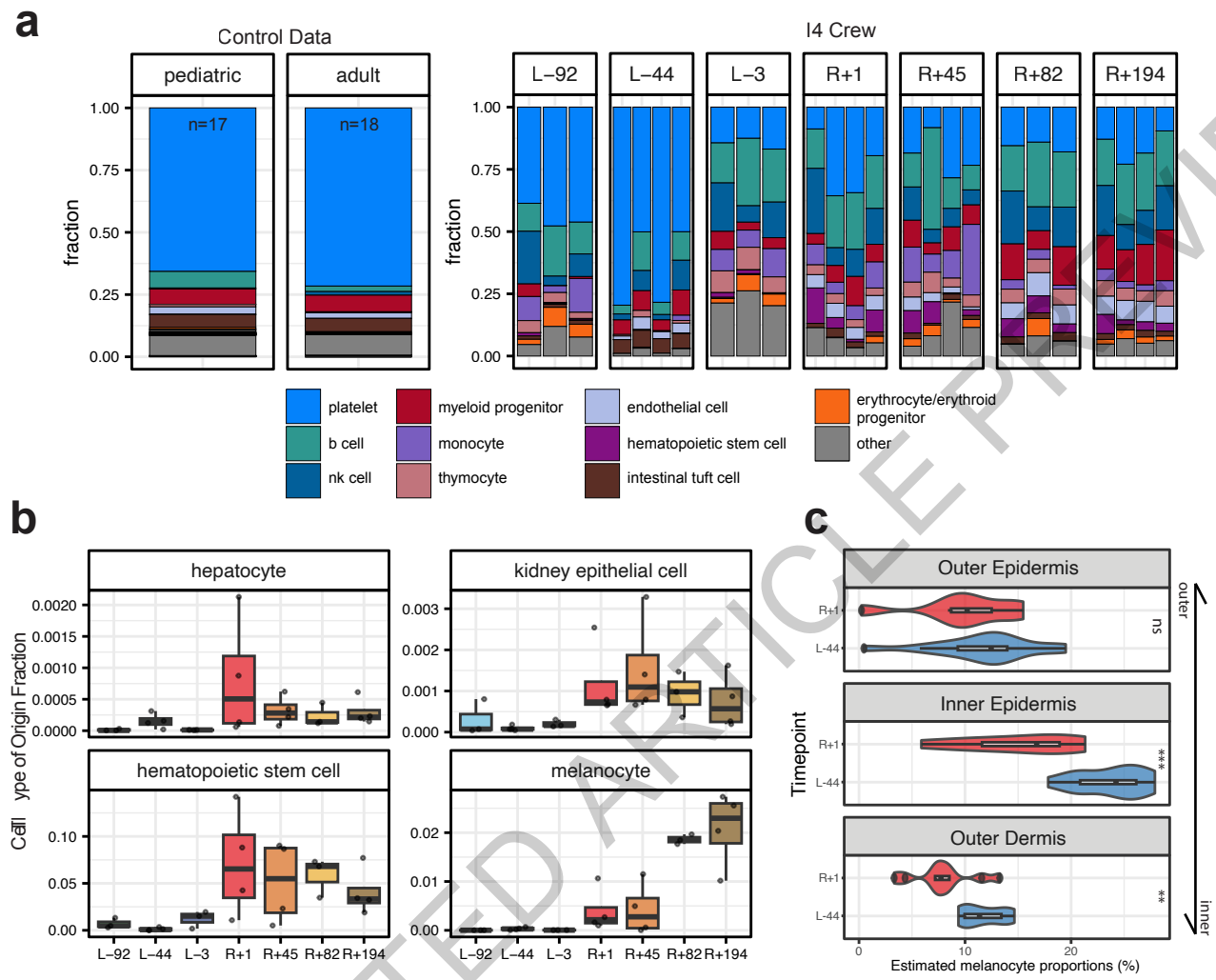

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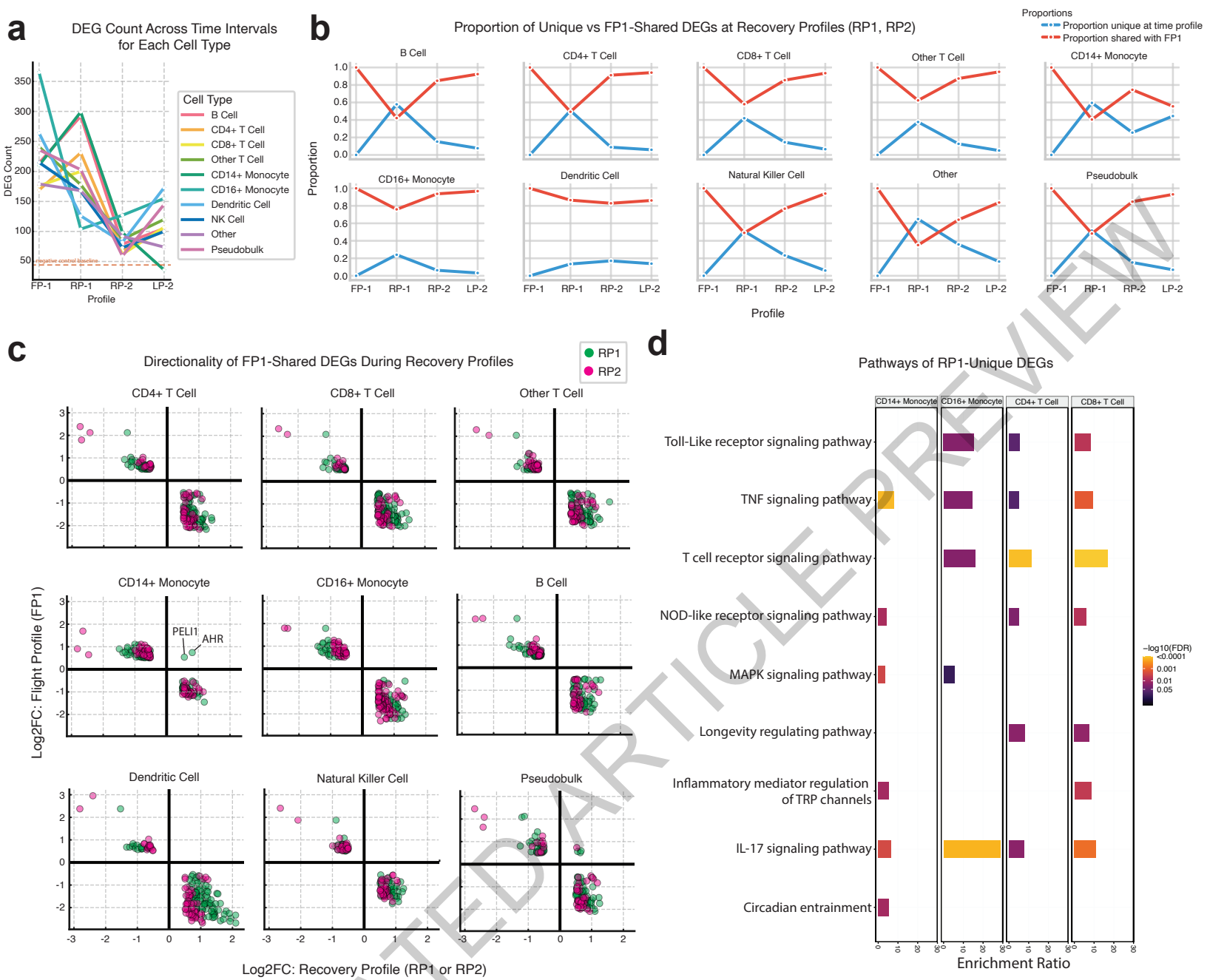

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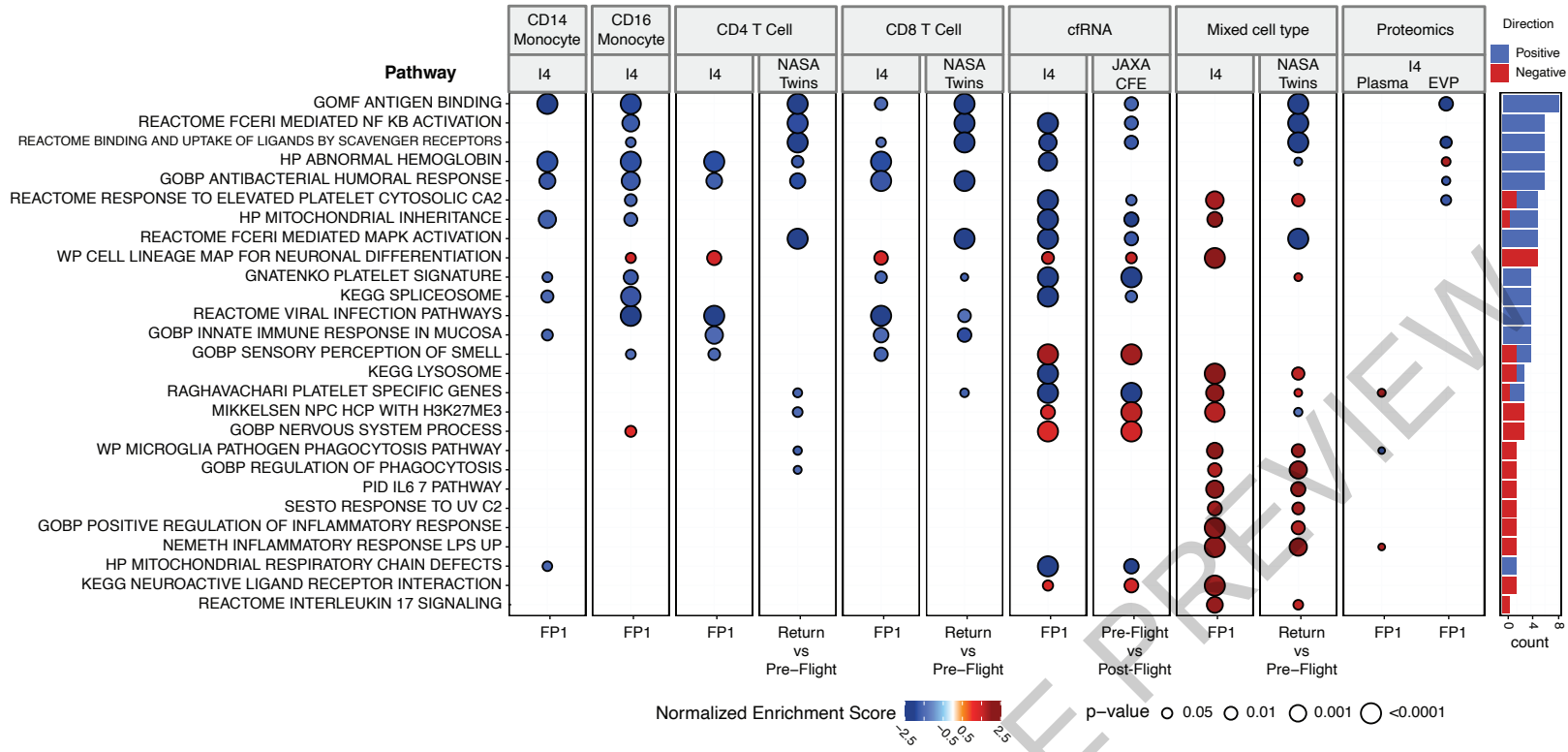

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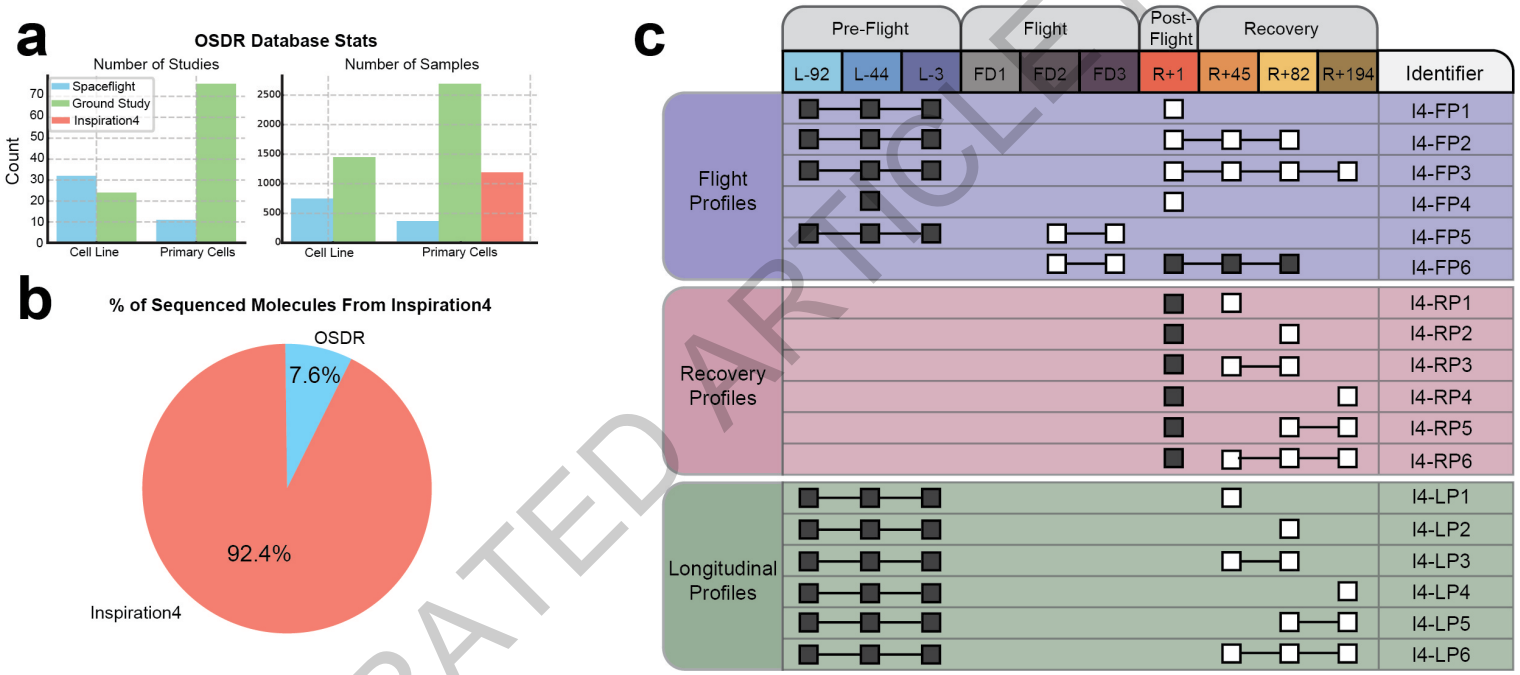

Extended Data Fig. 1

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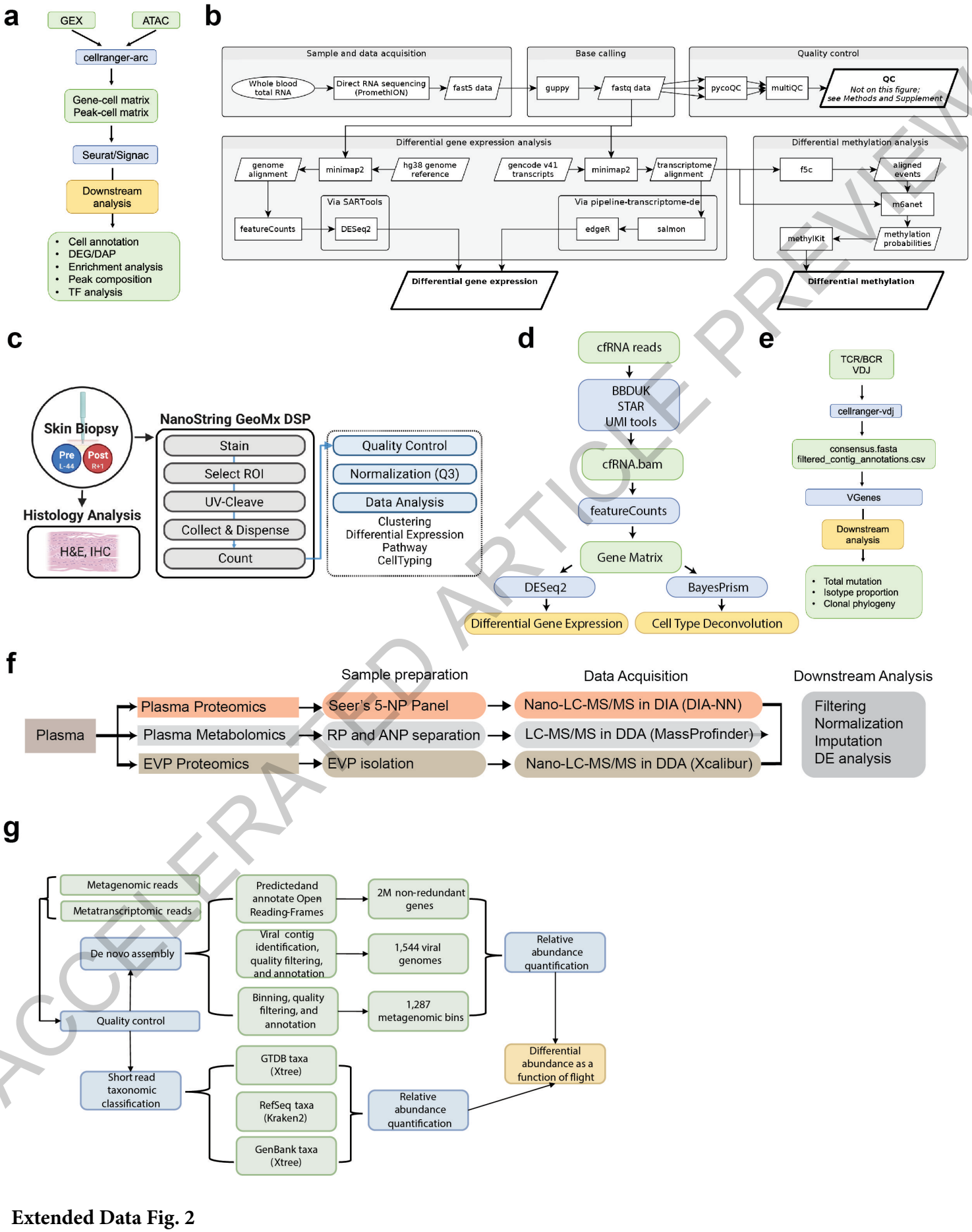

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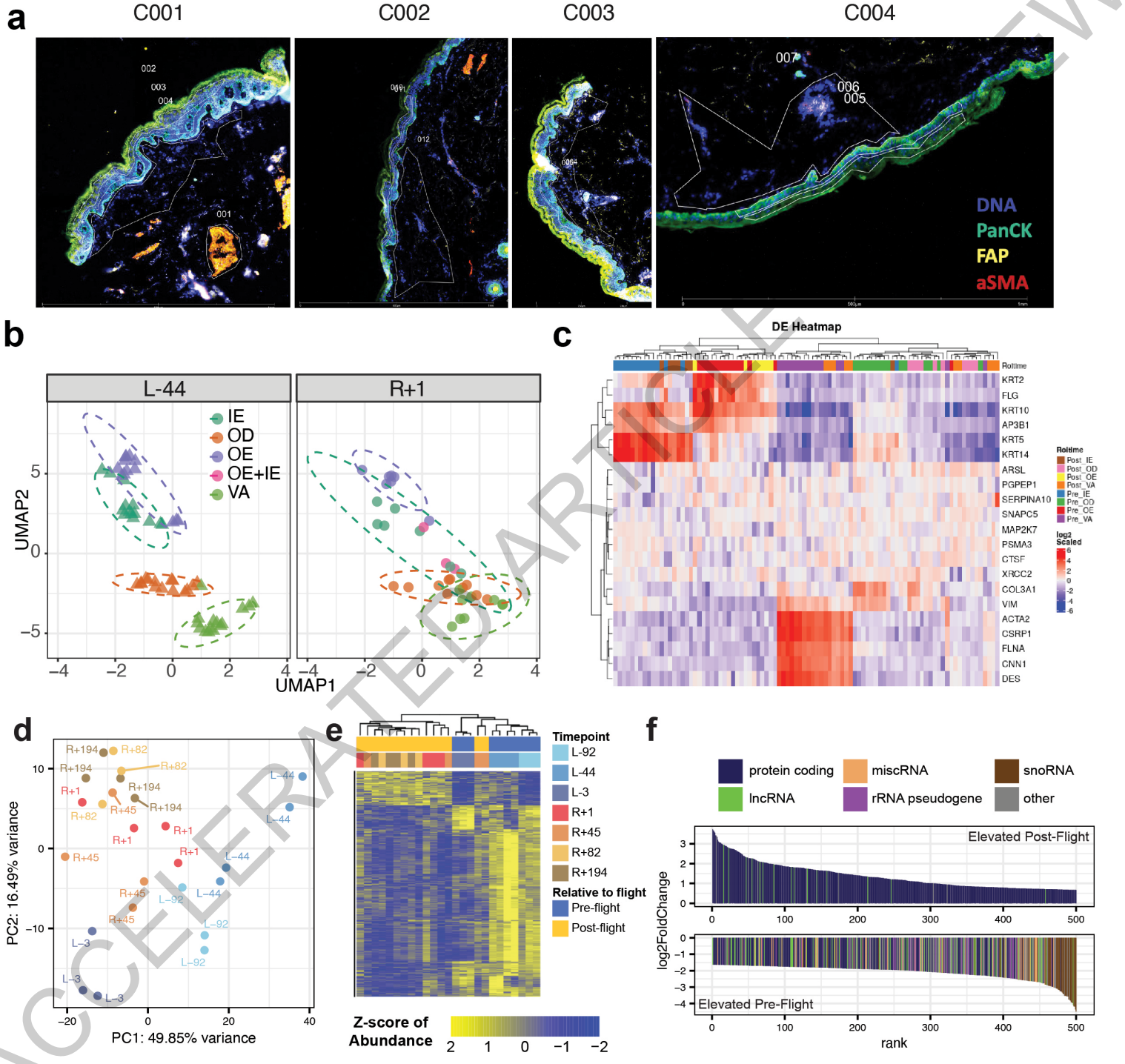

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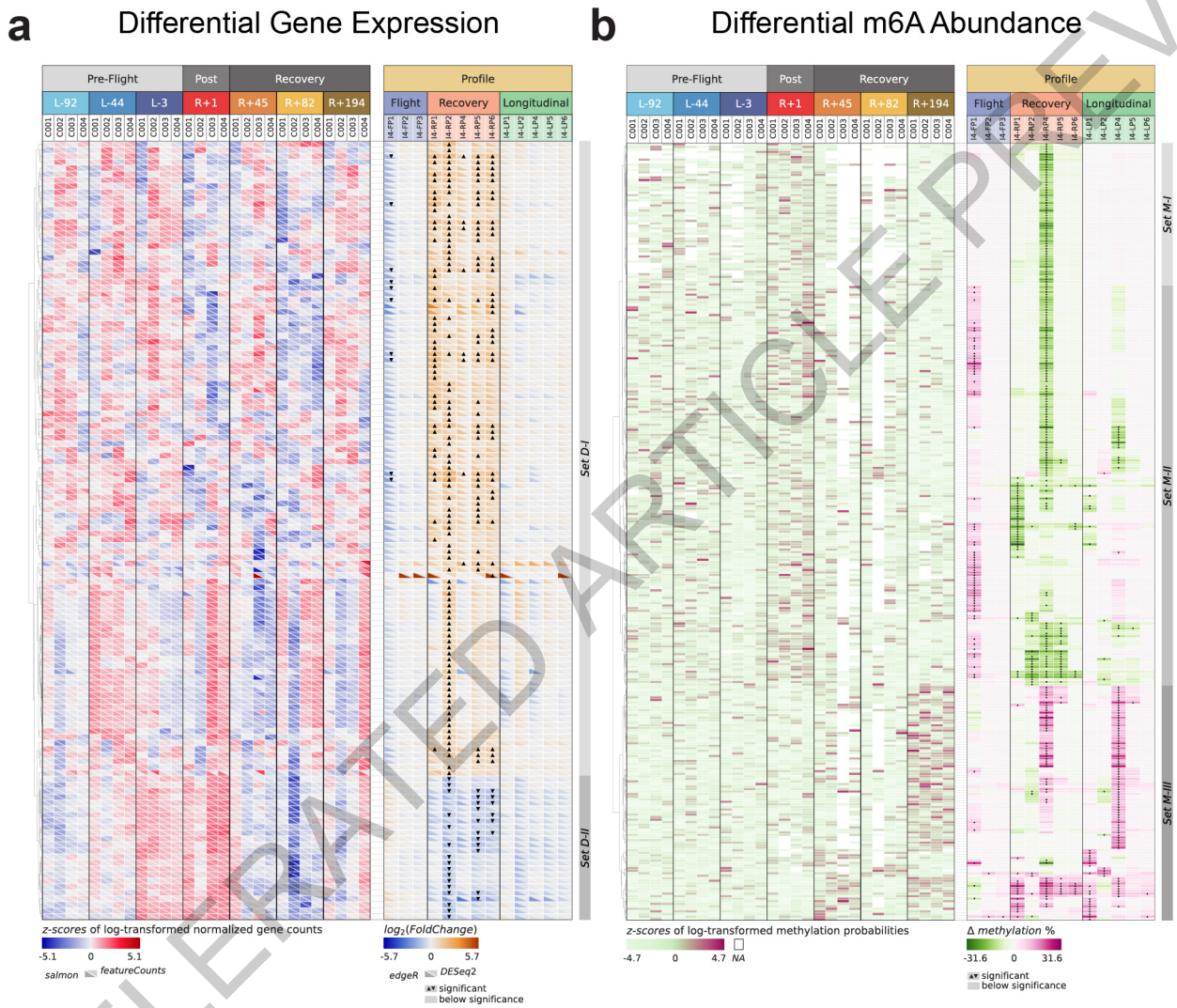

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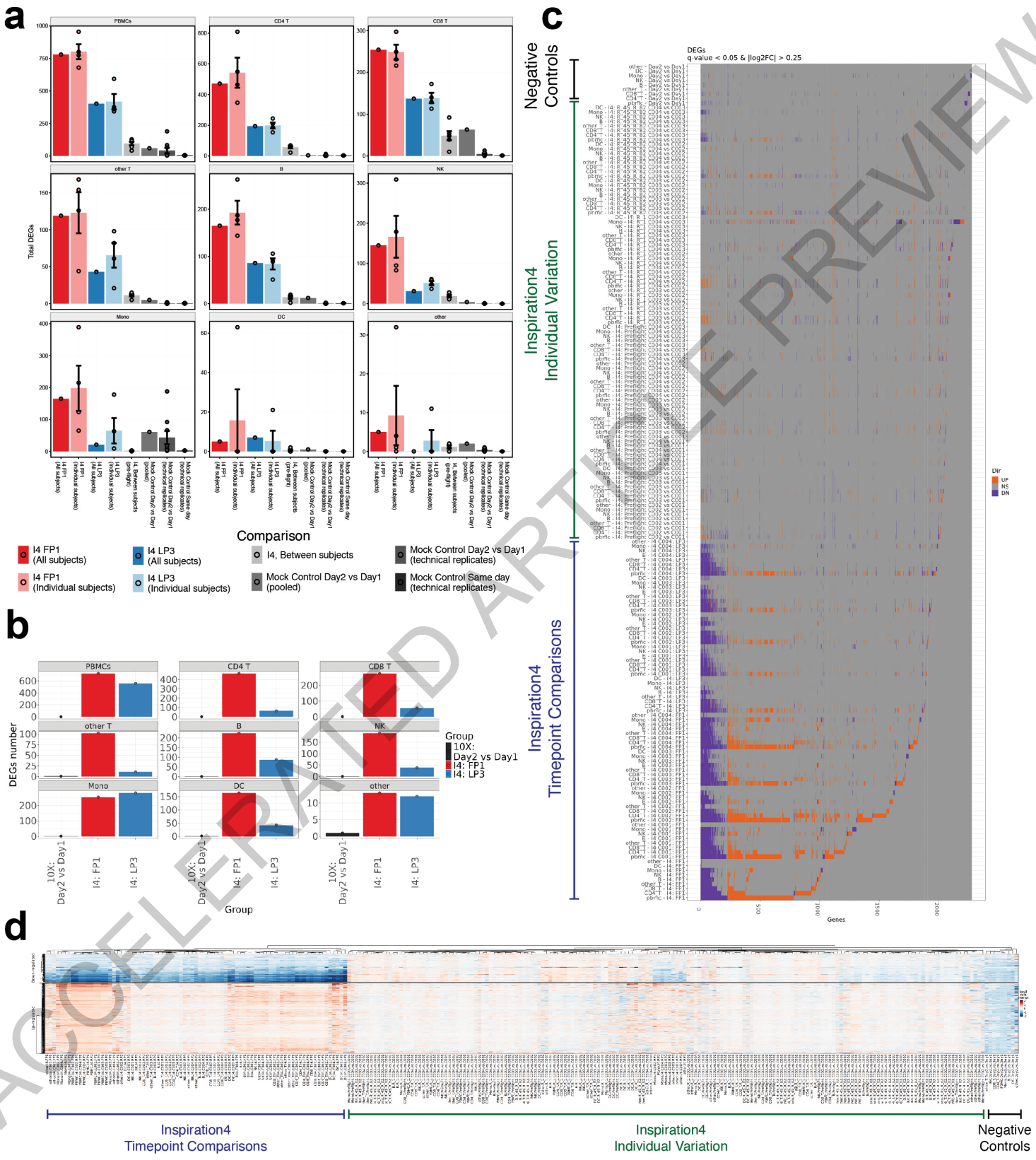

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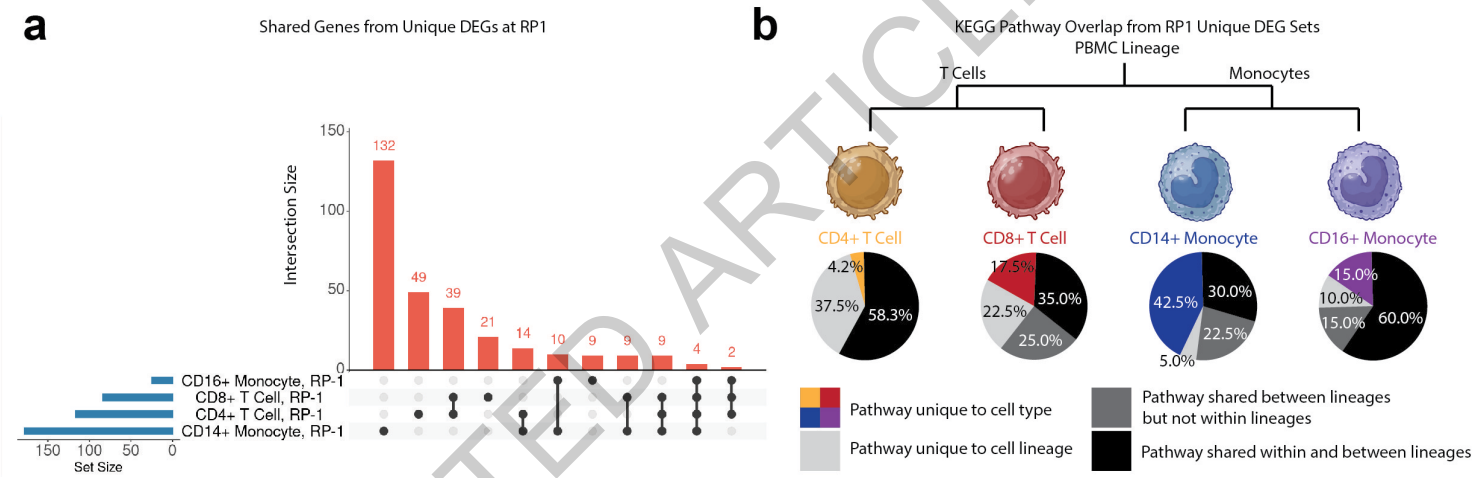

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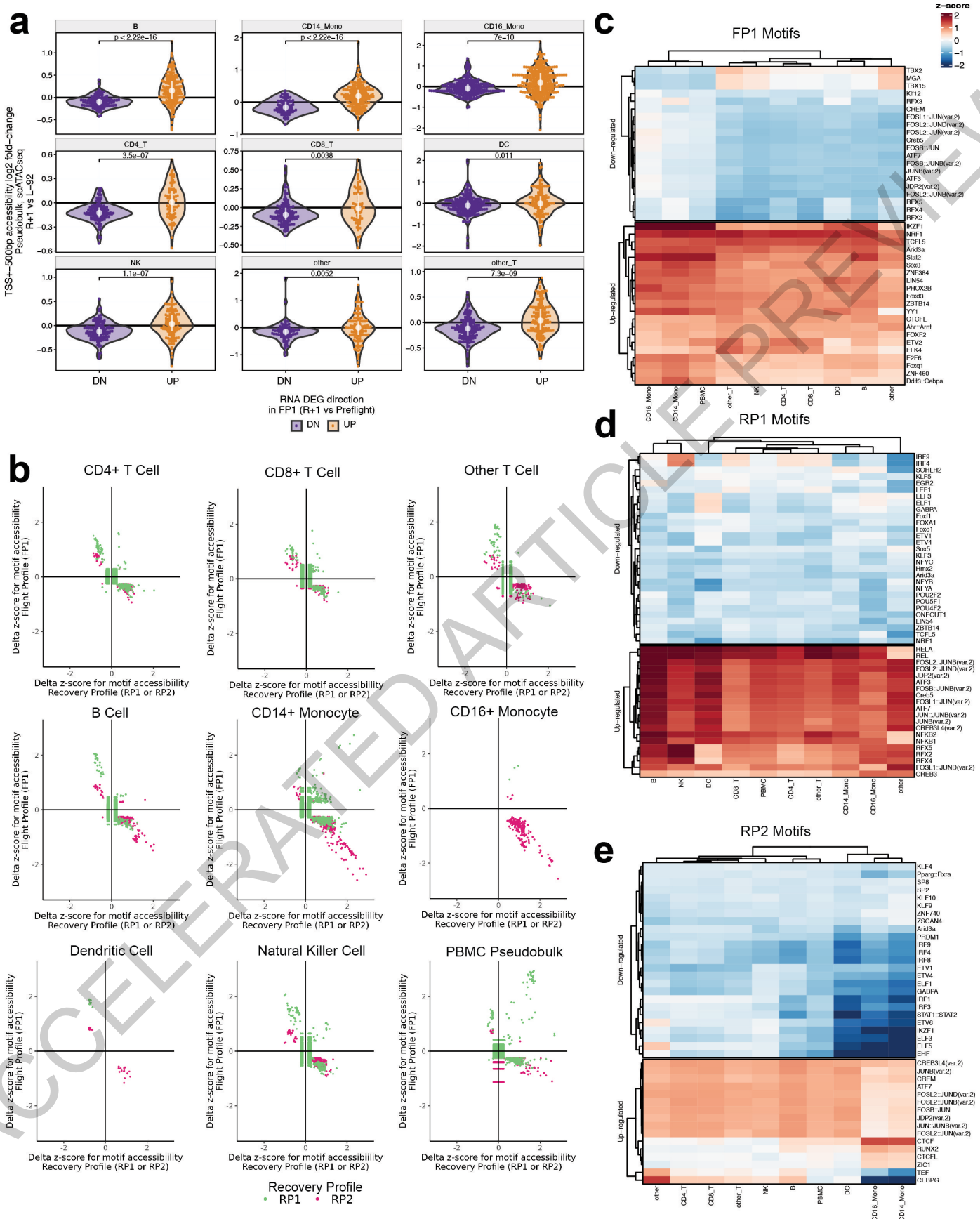

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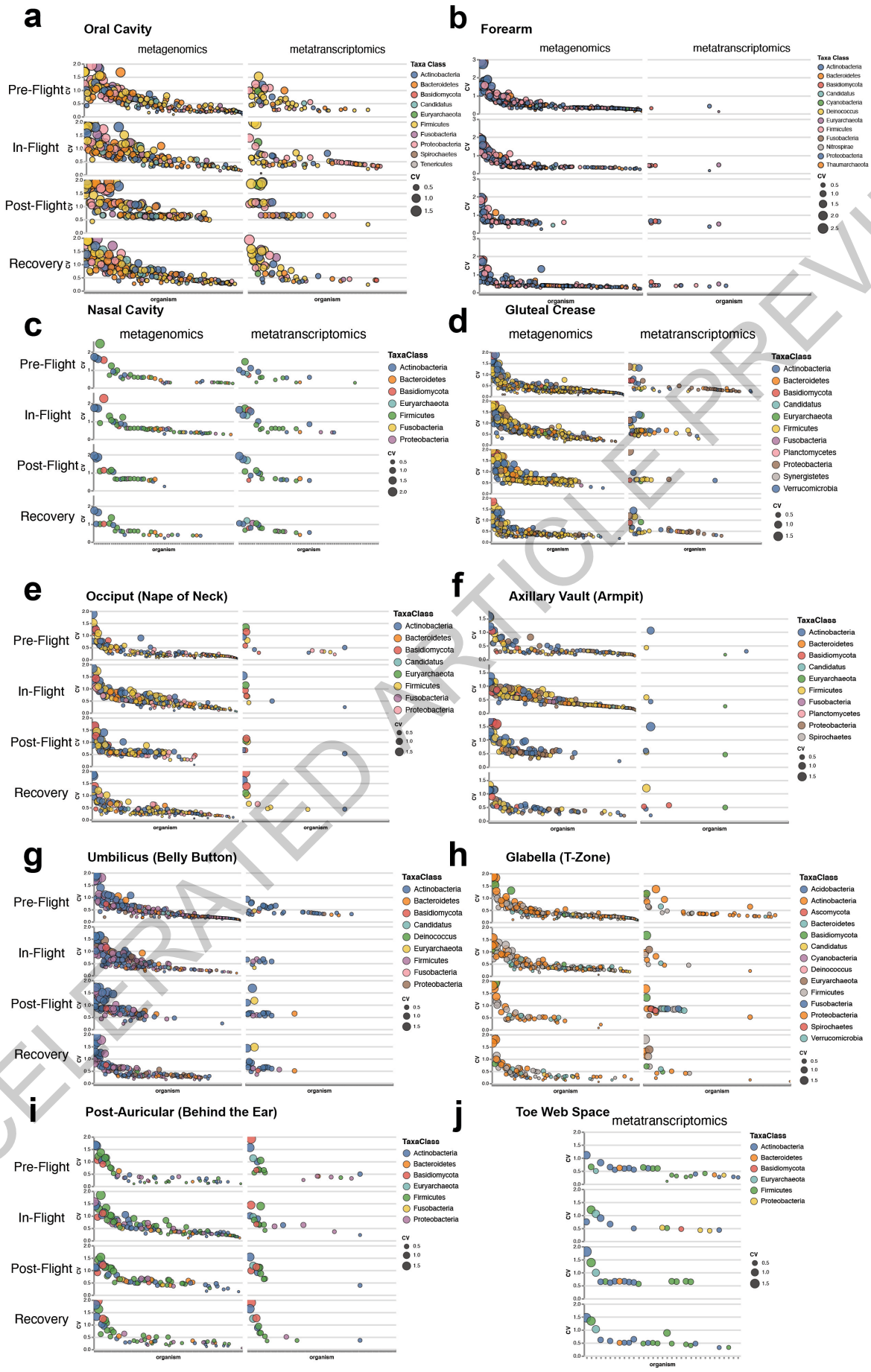

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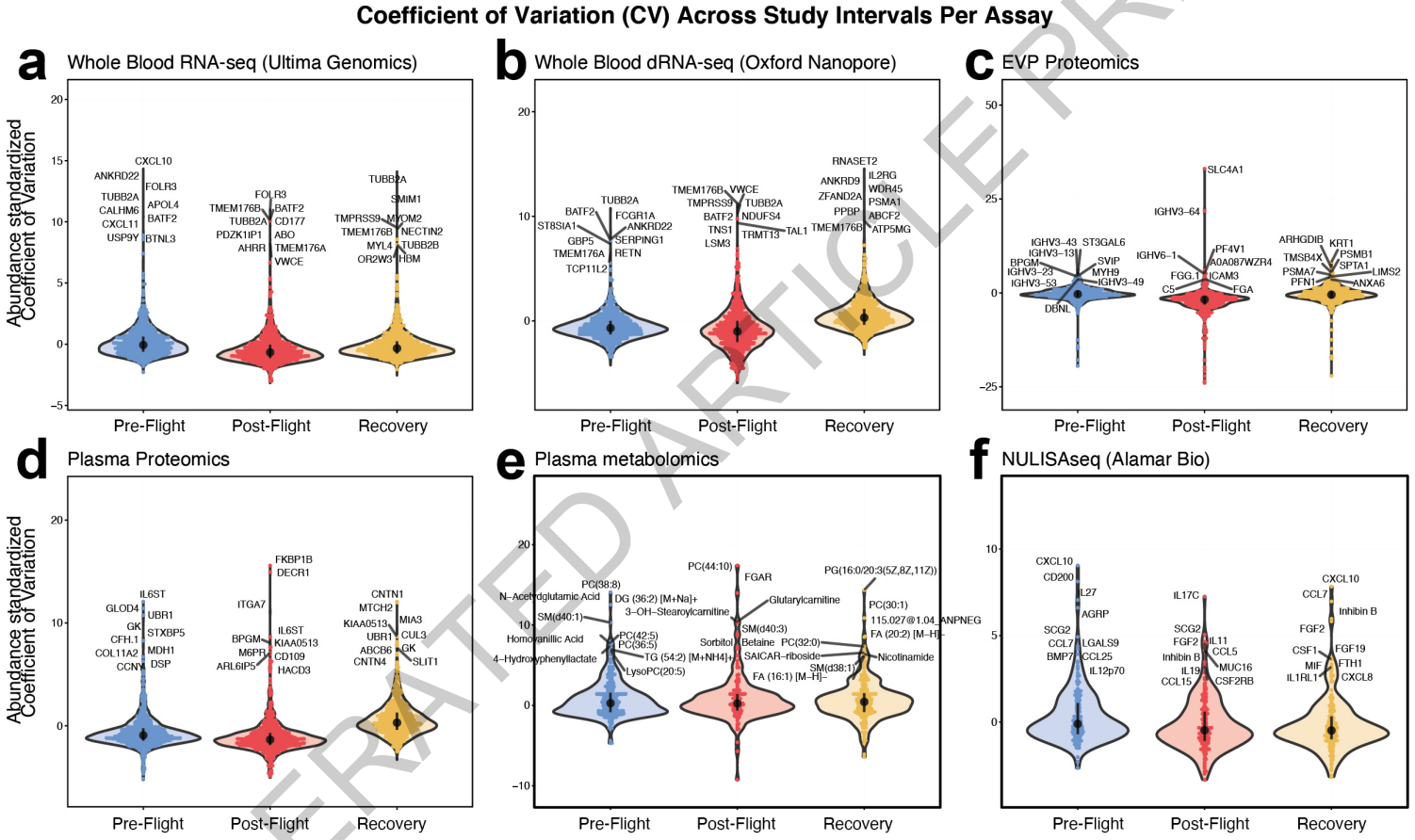

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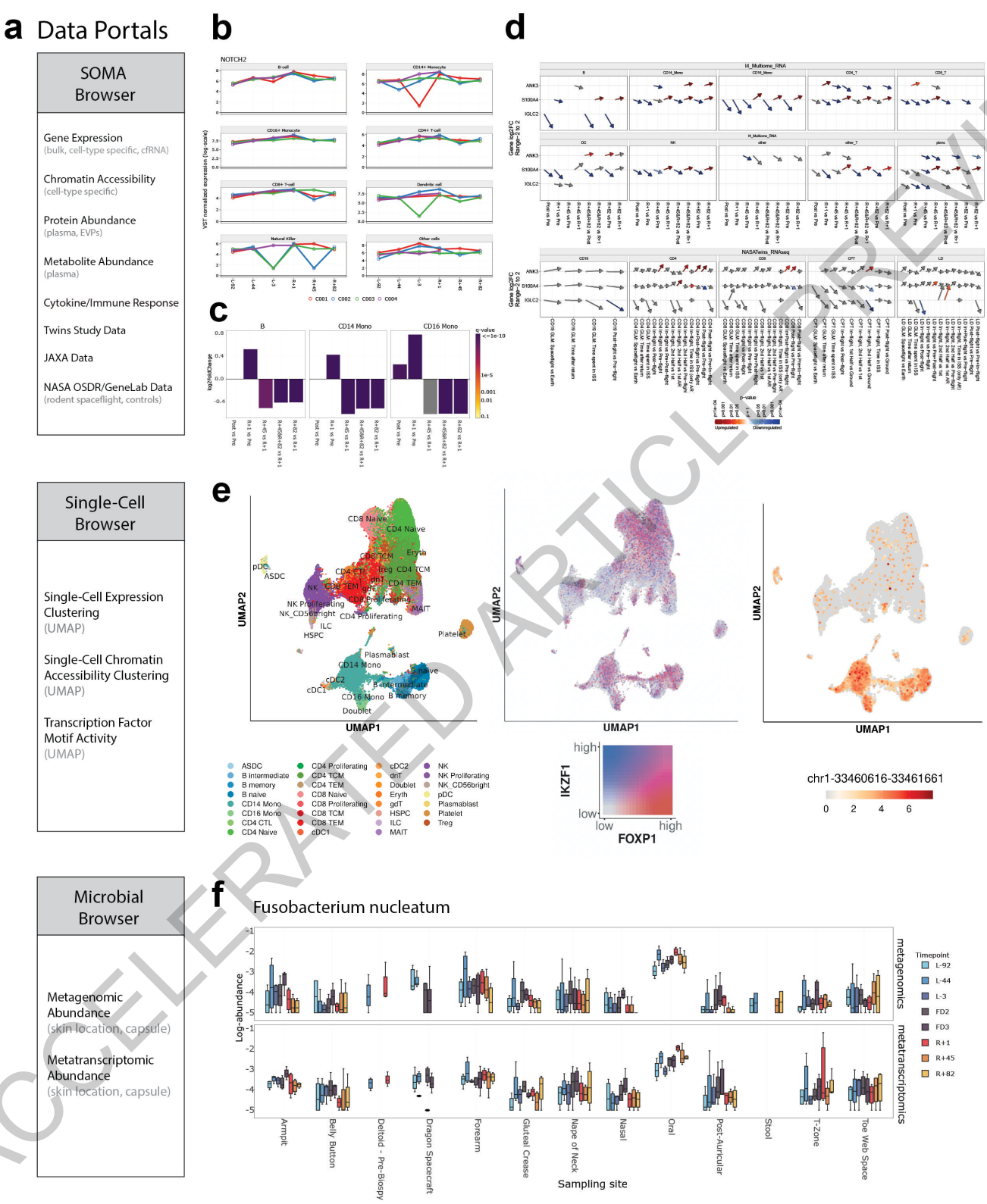

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Extended Data Fig. 10

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☒	☐	Estimates of effect sizes (e.g. Cohen's *d*, Pearson's *r*), indicating how they were calculated

Our web collection on statistics for biologists contains articles on many of the points above.

Software and code

Policy information about availability of computer code

Data collection	No software was used for data collection
Data analysis	ani-cluster (https://github.com/snayfach/MGV/tree/master/ani_cluster) Bakta (v1.5.1) bbtools (v38.92) BCFtools (v1.9) bedtools (v2.29.2) Bowtie2 (v2.2.3) bracken (v2.6.2) BWA MEM (v0.7.15) Cell Ranger (v6.1.1) CheckV (v0.8.1) CheckM (v1.2) deepTools (v3.5.1) DESeq2 (v1.36.0) Diamond (v2.0.14) FastQC (v0.11.9) fgsea (v1.22) GSVA (v1.42.0) HOMER (v4.11) Kraken2 (v2.1.2)

limma (v3.52)
m6Anet (v1.1.1)
MACS2 (v2.2.6)
MassProfinder 8.0
MetaQUAST (v5.0.2)
MetaSPAdes (v3.14.3)
methylKit (v3.18)
minimap2 (v2.24)
MMseqs2 (v13.4511)
MultiQC (v1.13)
pheatmap (v1.0)
PycoQC (v2.5.0.21)
samtools (v1.0, v1.9)
SARTools (v1.8.1)
SEACR (v1.3)
Sentieon TNscope RNA-seq variant pipeline (v202010)
Seurat (v4.2.0)
SnpEff (v4.3)
subread (v2.0.1)
trimmomatic (v0.39)
Varient-Effect-Predictor VEP (v107)
vegan (v2.6.2)
VGenes (vPre-release3)
vironomy (https://github.com/b-tierney/vironomy)
XTree (v0.92i)

Additional codes and scripts can be found in https://github.com/eliah-o/inspiration4-omics

For manuscripts utilizing custom algorithms or software that are central to the research but not yet described in published literature, software must be made available to editors and reviewers. We strongly encourage code deposition in a community repository (e.g. GitHub). See the Nature Portfolio guidelines for submitting code & software for further information.

Data

Policy information about availability of data

All manuscripts must include a data availability statement. This statement should provide the following information, where applicable:

- Accession codes, unique identifiers, or web links for publicly available datasets
- A description of any restrictions on data availability
- For clinical datasets or third party data, please ensure that the statement adheres to our policy

Datasets have been uploaded to the NASA Open Science Data Repository (osdr.nasa.gov; comprised of NASA GeneLab and the NASA Ames Life Sciences Data Archive)

OSD-569
https://osdr.nasa.gov/bio/repo/data/studies/OSD-569/
OSD-570
https://osdr.nasa.gov/bio/repo/data/studies/OSD-570/
OSD-571
https://osdr.nasa.gov/bio/repo/data/studies/OSD-571/
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https://osdr.nasa.gov/bio/repo/data/studies/OSD-575/
OSD-630
https://osdr.nasa.gov/bio/repo/data/studies/OSD-630/
OSD-656
https://osdr.nasa.gov/bio/repo/data/studies/OSD-656/

In addition the JAXA Cell-Free Epigenome (CFE) Study data and results were obtained from accession OSD-530.
https://osdr.nasa.gov/bio/repo/data/studies/OSD-530/

Processed data are also available for visualization in the following web portals.
https://soma.weill.cornell.edu/apps/SOMA_Browser/
https://soma.weill.cornell.edu/apps/I4_Multiome/
https://soma.weill.cornell.edu/apps/I4_Microbiome/

Human research participants

Policy information about studies involving human research participants and Sex and Gender in Research.

Reporting on sex and gender	Sex information was collected and annotated in the NASA OSDR repository for each sample collected.
Population characteristics	The crew member composition was of two races and ages ranged from 29-51.
Recruitment	Participants were recruited by SpaceX and mission commander Jared Isaacman through various task-specific criteria and raffle processes. There are no self-selection biases relevant to sample the collection protocols outlined in this paper.
Ethics oversight	All subjects were consented at an informed consent briefing (ICB) at SpaceX (Hawthorne, CA), and samples were collected and processed under the approval of the Institutional Review Board (IRB) at Weill Cornell Medicine, under Protocol 21-05023569. All crew members have consented for data and sample sharing.

Note that full information on the approval of the study protocol must also be provided in the manuscript.

Field-specific reporting

Please select the one below that is the best fit for your research. If you are not sure, read the appropriate sections before making your selection.

☒ Life sciences ☐ Behavioural & social sciences ☐ Ecological, evolutionary & environmental sciences

For a reference copy of the document with all sections, see nature.com/documents/nr-reporting-summary-flat.pdf

Life sciences study design

All studies must disclose on these points even when the disclosure is negative.

Sample size	No sample size calculations were performed. The entire Inspiration4 crew was profiled, which was limited by the size of the Dragon capsule (n=4).
Data exclusions	No data has been excluded.
Replication	Replication tests are difficult as mission parameters cannot be repeated. Where possible, data validation was performed via western blots to validate proteomic findings. Western blots were done only once due to limited material.
Randomization	No randomizations were performed, all of the subjects were profiled longitudinally and there were no experimental groups or interventions that separated the subjects.
Blinding	Blinding was not possible because all subjects were astronauts in the same crew.

Reporting for specific materials, systems and methods

We require information from authors about some types of materials, experimental systems and methods used in many studies. Here, indicate whether each material, system or method listed is relevant to your study. If you are not sure if a list item applies to your research, read the appropriate section before selecting a response.

Materials & experimental systems

n/a	Involved in the study
☐	☒ Antibodies
☒	☐ Eukaryotic cell lines
☒	☐ Palaeontology and archaeology
☒	☐ Animals and other organisms
☒	☐ Clinical data
☒	☐ Dual use research of concern

Methods

n/a	Involved in the study
☒	☐ ChIP-seq
☒	☐ Flow cytometry
☒	☐ MRI-based neuroimaging

Antibodies

Antibodies used	Abcam H3K27ac (CAT:ab245911) Epicypher H3K4me1 (SKU:13-0040)

Cutana Rabbit IgG antibody (SKU: 13-0042)

Cell Signaling, human CD9 (rabbit monoclonal IgG, CAT:13174S)
Santa Cruz Biotechnology, Galectin 3 Binding Protein (LGALS3BP) (mouse monoclonal IgG, CAT:sc-374541)
Abcam, Fibronectin (FN1) (rabbit polyclonal IgG, CAT:ab2413)
Cell Signaling, β-actin (rabbit monoclonal IgG, CAT:8457S)
Jackson Laboratory, horseradish peroxidase (HRP)-labeled IgG goat anti-rabbit or goat anti-mouse

Validation

H3K27ac and H3K4me1 antibodies were compared to the Cutana Rabbit IgG antibody, which served as a negative control.

Information from the manufacturer:
Abcam H3K27ac (CAT:ab245911)
This antibody meets the Abcam promise guarantee for immunocytochemistry and immunofluorescence applications. and manufactrer notes it specifically binds to K27ac alone and also when S28 is phosphorylated.

Epicypher H3K4me1 (SKU:13-0040)
This antibody meets EpiCypher's "SNAP-ChIP® Certified" criteria for specificity and efficient target enrichment in a ChIP experiment (<20% cross-reactivity across the panel, >5% recovery of target input). This antibody binds to H3K4me1 and no significant cross reactivity with other lysine methylations in the EpiCypher SNAP-ChIP K-MetStat Panel (EpiCypher 19-1001) is detected.

Cell Signaling human CD9 antibody is noted as recognizing endogenous levels of total human CD9 protein and certified for western blot use by the manufacturer.

Santa Cruz Biotechnology Galectin 3 Binding Protein antibody is noted as being suitable for detection of Galecting-3BP/Mac-2BP of mouse, rat and human origin by western blot use by the manufacturer.

Abcam Fibronectin antibody is covered by the Abcam guarantee for use in Western blot and the manufacturer notes that it detects a band of approximately 285 kDa (predicted molecular weight: 262 kDa).

Cell Signaling β-actin antibody is marked for Western blot use, and the manufacturer notes that iut recognizes endogenous levels of total β-actin protein. Due to the high sequence identity between the cytoplasmic actin isoforms, β-actin and cytoplasmic γ-actin, this antibody may cross-react with cytoplasmic γ-actin. It does not cross-react with α-skeletal, α-cardiac, α-vascular smooth, or γ-enteric smooth muscle isoforms. Species Reactivity: Human, Mouse, Rat, Monkey, D. melanogaster, Zebrafish